

Fine-Grained Classification of Unknown Fermi-LAT 4FGL-DR4 Sources with Spectral and Uncertainty Information

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ABSTRACT

We present an uncertainty-aware, fine-grained multiclass classification of unresolved *Fermi* Large Area Telescope (LAT) γ -ray sources in the 4FGL-DR4 catalog. We define the unresolved-source population as blank-class sources, sources labeled **unk**, and blazar candidates of uncertain type, and classify them within a supervised 22-class catalog taxonomy. The baseline analysis represents each source through the three catalog spectral forms—PowerLaw (PL), LogParabola (LP), and PLSuper-ExpCutoff (PLEC)—and compares TabNet, Random Forest, and a stacked TabNet+Random-Forest ensemble. The ensemble reaches accuracies of 92.95%, 92.45%, and 94.13% for the PL, LP, and PLEC representations, respectively. To quantify the role of feature choice, we further perform controlled ablations using an all-spectral configuration (AllSpec), which combines PL, LP, and PLEC parameters simultaneously, together with LP-only, PL-only, PLEC-only, and photometry-plus-variability configurations. Each configuration is tested with and without uncertainty columns and with $\lambda_{\max} = 0.1$ – 0.5 for synthetic interpolation. The strongest ablation result is obtained by the uncertainty-aware PLEC configuration, with accuracy and weighted F_1 of 0.9614. We also provide probabilistic predictions for all 4186 unresolved sources, including top-three class probabilities, full class-probability vectors, top-1–top-2 probability margins, and predictive entropy.

Keywords: Celestial objects catalogs (212) — Gamma-ray sources (633) — High energy astrophysics (739) — Classification (1907) — Algorithms (1883)

1. INTRODUCTION

The *Fermi* Gamma-ray Space Telescope and its Large Area Telescope (*Fermi*-LAT) transformed GeV γ -ray astronomy by enabling sensitive, wide-field, long-term surveys of the high-energy sky (W. B. Atwood et al. 2009). Since launch, LAT observations have produced increasingly comprehensive source catalogs. The LAT catalog sequence began with 1FGL and 2FGL, continued through 3FGL, and then expanded into the 4FGL series and its data releases (A. A. Abdo et al. 2010a; P. L. Nolan et al. 2012; F. Acero et al. 2015; S. Abdollahi et al. 2020; J. Ballet et al. 2020; S. Abdollahi et al. 2022; J. Ballet et al. 2023). Using eight years of LAT observations, the *Fermi*-LAT Collaboration compiled the original 4FGL catalog and reported 5,064 sources (S. Abdollahi et al. 2020). In 4FGL-DR2 and 4FGL-DR3, the collaboration extended the time coverage to 10 and 12 yr, updated spectral and variability measurements, and refined

source associations (J. Ballet et al. 2020; S. Abdollahi et al. 2022). In 4FGL-DR4, the collaboration used 14 yr of LAT observations over 50 MeV–1 TeV and cataloged 7,195 γ -ray sources (J. Ballet et al. 2023). Many LAT sources still lack secure associations or detailed classifications, showing that source detection has outpaced physical identification.

Among the extragalactic populations detected by *Fermi*-LAT, active galactic nuclei (AGNs) dominate, and blazars account for most of them. In the unified AGN framework, blazars are radio-loud AGNs whose relativistic jets point close to our line of sight, producing strong Doppler boosting, rapid variability, and broadband nonthermal emission (C. M. Urry & P. Padovani 1995). Early optical-polarization studies supported the beaming interpretation of core-dominated radio quasars (B. J. Wills et al. 1992). Later work showed that variability, spectral properties, and jet physics separate blazar subclasses and encode their physical differences (J.-H. Fan 2002; J.-H. Fan et al. 2014; J. H. Fan et al. 2021; J.-H. Fan et al. 2022; J. Fan et al. 2023). AGNs also span a broader range of accretion and emission states than the classical blazar paradigm alone captures (X. W. Shu et al. 2017). The LAT AGN catalogs provide the population context for this picture, from 1LAC and 2LAC through 3LAC and the 4LAC releases (A. A. Abdo et al. 2010b; M. Ackermann et al. 2011, 2015; M. Ajello et al. 2020, 2022). Researchers usually divide blazars into BL Lacertae objects (BLLs) and flat-spectrum radio quasars (FSRQs), and they assign an intermediate group to the blazar candidates of uncertain type (BCU) category. The BCU label identifies a source as a blazar candidate but does not determine its final subclass. In practice, BCUs remain unresolved mainly at the subclass level, especially between BLL-like and FSRQ-like objects. Because blazars dominate the associated extragalactic LAT population, improved subclassification can sharpen blazar population studies, reduce ambiguity in the unresolved *Fermi*-LAT sky, and guide multiwavelength follow-up.

Machine learning now plays a central role in *Fermi*-LAT source studies (N. M. Ball & R. J. Brunner 2010; S. Sen et al. 2022). Instead of relying only on positional associations, ML methods use catalog observables, including spectral shape, curvature, flux distribution, and variability, to infer the likely physical classes of unresolved sources. Earlier statistical classification already established this direction. Gaussian-mixture modeling ranked pulsar candidates in the 2FGL catalog (K. J. Lee et al. 2012), and SIBYL used Random Forest to classify unassociated LAT sources at high Galactic latitude (N. Mirabal et al. 2012). Time-domain studies also showed that variability carries strong source-class information, as demonstrated by FAVA and the 2FAV catalog (M. Ackermann et al. 2013; S. Abdollahi et al. 2017). Later work focused mainly on binary or ternary tasks. P. M. Saz Parkinson et al. (2016) applied several statistical and machine-learning techniques to classify and rank 3FGL sources. G. Chiaro et al. (2016) used blazar flaring patterns and an artificial neural network to classify BCUs in 3FGL (F. Acero et al. 2015). D. Salvetti et al. (2017) applied artificial neural networks to classify 3FGL unassociated sources in a broader source-identification setting. H. Xiao et al. (2020) applied several ML classifiers and an ensemble strategy to identify AGNs among unassociated 3FGL sources and to separate BLLs from FSRQs within the BCU sample. Using 4FGL data, J. T. Zhu et al. (2021) distinguished AGN and pulsar candidates among unassociated sources. A. Bhat & D. Malyshev (2022) built probabilistic two- and three-class catalogs, K. Cooper et al. (2023) combined missing-value imputation with SuperLearner-based ensemble classification for AGN subclassification, K. R. Zhu et al. (2024) analyzed unassociated 4FGL-DR3 sources with multiple ML methods, and A. Tolamatti et al. (2023) used XGBoost to classify BCUs. These studies showed that LAT catalog observables carry substantial class information. However, most addressed binary, ternary, or coarsely grouped tasks rather than direct fine-grained catalog-level subclassification (G. Chiaro et al. 2016; H. Xiao et al. 2020; J. T. Zhu et al. 2021; A. Bhat & D. Malyshev 2022; K. Cooper et al. 2023; A. Tolamatti et al. 2023; S. Germani et al. 2021; D. Malyshev & A. Bhat 2023; D. V. Malyshev 2023; J. Coronado-Blázquez 2022). In the specific problem of BCU identification, both neural-network and boosting-based approaches have shown that catalog-level observables can recover useful subclass information even when the final optical classification is unavailable (G. Chiaro et al. 2016; G. Chiaro et al. 2019; A. Tolamatti et al. 2023; T. Finke et al. 2021). More broadly, machine learning in astronomy has expanded rapidly across both supervised and unsupervised settings, including source classification, anomaly detection, and representation learning (N. M. Ball & R. J. Brunner 2010; D. Baron 2019; S. Sen et al. 2022; S. G. Djorgovski et al. 2022; S. Fotopoulou et al. 2024).

Recent work has moved beyond binary and ternary classification toward genuine multiclass classification of unresolved *Fermi*-LAT sources. S. Germani et al. (2021) performed a three-category classification of 4FGL DR1 and DR2 sources with an ensemble of artificial neural networks and assigned sources to PSR, BLL, and FSRQ classes. D. Malyshev & A. Bhat (2023) introduced a hierarchical multiclass framework for 4FGL-DR3 and showed that grouping related classes improves classification stability under severe class imbalance. D. V. Malyshev (2023) examined covariate shift between associated and unassociated samples and developed probabilistic catalogs in a multiclass setting. Together, these

studies show that the unresolved LAT population is intrinsically heterogeneous and that fine-grained classification is both feasible and astrophysically useful.

We pursue a more detailed classification of the unknown-source population in 4FGL-DR4. We include sources labeled as “unk,” sources with blank class labels, and BCUs. We make this choice because the BCU label identifies a broad blazar type but does not resolve a source into a specific subclass, such as BLL or FSRQ. We therefore treat BCUs as subclass-unknown sources rather than as fully identified sources. Under this definition, we formulate the problem as a fine-grained classification task over 22 catalog classes. We divide the data into the three spectral-shape types defined in 4FGL-DR4: LogParabola (LP), PowerLaw (PL), and PLSuperExpCutoff (PLEC). We then build an independent classification pipeline for each spectral representation. We compare three complementary methods: TabNet, a deep-learning model for tabular data with sequential attentive feature selection (S. O. Arik & T. Pfister 2021); Random Forest (RF), a robust tree-based model widely used for structured astronomical data (L. Breiman 2001); and an ensemble model that combines TabNet and RF predictions. Unlike previous studies that treated the catalog feature space as fixed, we evaluate whether the adopted spectral representation changes multiclass classification performance and source-level subclass assignments. By combining spectral-type-aware modeling, a fine-grained taxonomy, and hybrid learning, we derive more detailed catalog-level subclass predictions for unknown 4FGL-DR4 sources in a setting complementary to recent probabilistic, hierarchical, and direct multiclass studies of LAT catalogs (A. Bhat & D. Malyshev 2022; S. Germani et al. 2021; D. Malyshev & A. Bhat 2023; D. V. Malyshev 2023; J. Coronado-Blázquez 2022).

This work keeps the formal PL, LP, and PLEC spectral-representation framework and the comparison among TabNet, RF, and the stacked ensemble, but uses the expanded experiments to determine which feature configuration is most reliable. We test an AllSpec configuration, in which PL, LP, and PLEC parameters are used simultaneously, together with LP-only, PL-only, PLEC-only, and photometry-plus-variability configurations. This design measures whether the full catalog spectral information improves over single-representation pipelines and whether non-spectral photometric and variability information alone can sustain strong classification. We also compare uncertainty-aware and uncertainty-free versions of each configuration, treating uncertainty columns as measurement-quality descriptors rather than physical class labels. Finally, the unresolved-source catalog is reported probabilistically: for each source we provide the top-three class probabilities, the full 22-class probability vector, the top-1–top-2 probability margin, and predictive entropy.

We organize the rest of this paper as follows. In Sec. 2, we describe the data sample and define the unknown-source set. In Sec. 3, we describe the model architecture, hyperparameters, and performance metrics. In Sec. 4, we present the classification results. In Sec. 5, we discuss the implications and summarize our conclusions.

2. SAMPLE CONSTRUCTION AND DATA DESCRIPTION

2.1. The 4FGL-DR4 Catalog

The fourth *Fermi* Large Area Telescope source catalog Data Release 4 (4FGL-DR4) provides the observational basis of this work. The 4FGL series began with the 8-year catalog and expanded through DR2, DR3, and DR4 as LAT exposure increased and analysis improved (S. Abdollahi et al. 2020; J. Ballet et al. 2020; S. Abdollahi et al. 2022; J. Ballet et al. 2023). As the latest large-scale census of the GeV γ -ray sky, 4FGL-DR4 provides a uniform set of positional, spectral, and temporal quantities for each detected source, including band fluxes, spectral parameters, spectral-energy-distribution descriptors, and variability measurements.

The LAT source population spans a broad range of Galactic and extragalactic objects. In this work, the labeled sample contains 22 catalog classes: `agn`, `bin`, `bll`, `css`, `fsrq`, `gal`, `gc`, `glc`, `hmb`, `lmb`, `msp`, `nlsy1`, `nov`, `psr`, `pwn`, `rdg`, `sbg`, `sey`, `sfr`, `snr`, `spp`, and `ssrq`. We use these labeled catalog classes as the target space for supervised classification and then apply the trained models to the unresolved-source sample. We therefore study a fine-grained catalog-classification problem rather than a coarse binary task.

Table 1 defines all 22 catalog class labels used in the supervised target space. In this table, Col. (1) gives the lower-case 4FGL-DR4 class code used as the supervised label, Col. (2) expands the code into a representative astrophysical meaning, and Col. (3) assigns the label to a broad source family for orientation. These definitions are included to make the catalog taxonomy self-contained.

Table 1. Definitions of the 22 catalog class labels used in this work.

Class code	Representative meaning	Broad group
agn	Non-blazar active galactic nucleus	Extragalactic AGN
bin	Binary system	Galactic binary/transient
bll	BL Lacertae object	Blazar/AGN
css	Compact steep-spectrum radio source	Extragalactic AGN
fsrq	Flat-spectrum radio quasar	Blazar/AGN
gal	Normal galaxy or galaxy-related source	Extragalactic
gc	Galactic center source	Galactic system
glc	Globular cluster	Galactic system
hmb	High-mass binary	Galactic binary/transient
lmb	Low-mass binary	Galactic binary/transient
msp	Millisecond pulsar	Galactic compact object
nlsy1	Narrow-line Seyfert 1 galaxy	Extragalactic AGN
nov	Nova	Galactic transient
psr	Young pulsar	Galactic compact object
pwn	Pulsar-wind nebula	Galactic extended source
rdg	Radio galaxy	Extragalactic AGN
sbg	Starburst galaxy	Extragalactic galaxy
sey	Seyfert galaxy	Extragalactic AGN
sfr	Star-forming region or star-forming-galaxy-related source	Galactic/extragalactic star-forming system
snr	Supernova remnant	Galactic extended source
spp	Supernova-remnant/pulsar-wind-nebula association	Galactic extended or confused source
ssrq	Steep-spectrum radio quasar	Extragalactic AGN

The supervised target space follows the 4FGL-DR4 catalog labels. The **bcu** and **unk** labels are not used as final target classes; they are included in the unresolved-source sample to be classified. Rare classes with very small validation support are retained for catalog completeness but are interpreted with caution.

2.2. Definition of the Unknown-Source Sample

We define the unknown-source sample as the union of sources with blank class labels, sources labeled as **unk**, and blazar candidates of uncertain type (**bcu**). We do not treat **bcu** as a final astrophysical subclass. Instead, we treat it as an unresolved source type with known broad blazar nature but uncertain subclassification, especially between BLL-like and FSRQ-like objects. This definition better reflects the physical ambiguity of these sources than a definition based solely on catalog incompleteness. It is consistent with the treatment of BCUs in LAT AGN catalogs as blazar candidates whose final subclassification remains open (M. Ajello et al. 2020, 2022; J. Ballet et al. 2023).

Using the processed catalog table, which has the same row count across the PL, LP, and PLEC representations, we find that the full sample contains 7,195 sources. Among them, 4,186 sources belong to the unknown-source population, including 2,423 blank-class sources, 1,623 BCUs, and 140 **unk** sources. The labeled sample contains the remaining 3,009 sources distributed across the 22 target classes.

The labeled sample is strongly imbalanced. The largest classes are **bll** (1490), **fsrq** (820), **msp** (179), **psr** (141), and **spp** (132), whereas several rare classes contain only a few sources, such as **sey** (4), **ssrq** (2), and **gc** (1). In the implemented workflow, the singleton **gc** label is retained in the supervised target space and therefore participates formally in training, validation, and prediction. However, because this class contains only one real labeled source, we do not interpret its class-specific validation performance as statistically reliable, even though we retain the label in training and prediction for catalog completeness. This imbalance is an intrinsic property of the classification problem.

2.3. Spectral Shapes in 4FGL-DR4

We use the three spectral shapes adopted by the catalog:

- **PowerLaw (PL)**,
- **LogParabola (LP)**,
- **PLSuperExpCutoff (PLEC)**.

The catalog defines the normalization of all three models through the flux density K at a reference energy E_0 , which it chooses to minimize the uncertainty in K . The catalog reports E_0 as `Pivot_Energy`. This choice reduces the correlation between the normalization and the shape parameters, providing a stable basis for spectral fitting (S. Abdollahi et al. 2020, 2022).

The **PowerLaw** model takes the form

$$\frac{dN}{dE} = K \left(\frac{E}{E_0} \right)^{-\Gamma}. \quad (1)$$

The catalog reports the corresponding parameters as `PL_Flux_Density` and `PL_Index`.

The **LogParabola** model takes the form

$$\frac{dN}{dE} = K \left(\frac{E}{E_0} \right)^{-\alpha - \beta \ln(E/E_0)}, \quad (2)$$

where the catalog reports the parameters as `LP_Flux_Density`, `LP_Index`, and `LP_beta`. The LP model also yields a peak-energy descriptor in νF_ν ,

$$E_{\text{peak}} = E_0 \exp \left(\frac{2 - \alpha}{2\beta} \right), \quad (3)$$

which the catalog reports as `LP_EPeak` (S. Abdollahi et al. 2020, 2022).

The **PLSuperExpCutoff** model describes spectra with cutoff-like behavior and is especially relevant for pulsar-like sources, which commonly show strong curvature or exponentially suppressed high-energy emission in LAT catalogs (A. A. Abdo et al. 2013; S. Abdollahi et al. 2022; J. Ballet et al. 2023). In the 4FGL description, the model takes the form

$$\frac{dN}{dE} = K \left(\frac{E}{E_0} \right)^{-\Gamma} \exp [a (E_0^b - E^b)], \quad (4)$$

where the catalog reports the parameters as `PLEC_Flux_Density`, `PLEC_Index`, `PLEC_Expfactor`, and `PLEC_Exp_Index`. In the DR3/DR4 analysis, the collaboration also introduced a more stable parameterization,

$$\frac{dN}{dE} = K \left(\frac{E}{E_0} \right)^{\frac{d}{b} - \Gamma_S} \exp \left[\frac{d}{b^2} \left(1 - \left(\frac{E}{E_0} \right)^b \right) \right], \quad (5)$$

in which Γ_S gives the local slope at E_0 and d gives the local curvature. The catalog reports these quantities as `PLEC_IndexS` and `PLEC_ExpfactorS`, respectively (S. Abdollahi et al. 2022).

In the limit $b \rightarrow 0$, the PLEC expression approaches the LP form, with

$$d = 2\beta. \quad (6)$$

Thus, Γ_S in PLEC plays a role analogous to that of α in LP, whereas d plays a role analogous to that of 2β . The corresponding peak energy in νF_ν is

$$E_{\text{peak}}(\text{PLEC}) = E_0 \left(1 + \frac{b}{d}(2 - \Gamma_S) \right)^{1/b}, \quad (7)$$

which the catalog reports as `PLEC_EPeak` (S. Abdollahi et al. 2022).

For clarity, we group the input quantities into five feature families. Photometric quantities include `Flux1000`, `Energy_Flux100`, `Flux_Band1–Flux_Band8`, `nuFnu_Band1–nuFnu_Band8`, and `Flux_Peak`. Variability quantities include `Variability_Index` and `Frac_Variability`. Spectral-shape quantities include the PL, LP, and PLEC model parameters listed in Table 6. The catalog `SpectrumType` field is treated as a coarse best-fit spectral-form descriptor and is one-hot encoded in the AllSpec, LP-only, PL-only, and PLEC-only ablation experiments. Uncertainty columns, including `Unc_*` quantities and band-flux uncertainties, are used only in uncertainty-aware configurations as measurement-quality descriptors. In particular, `PL_Flux_Density`, `LP_Flux_Density`, and `PLEC_Flux_Density` are spectral-model normalizations at the pivot energy rather than ordinary band-photometry features.

Each 4FGL-DR4 source has representations in all three spectral forms: PL, LP, and PLEC. These forms do not define three disjoint source populations. Instead, they provide three parallel spectral representations of the same catalog source set. The row count remains the same at the catalog level, whereas the feature columns change because the spectral parameters depend on the adopted model (S. Abdollahi et al. 2020, 2022). We therefore classify and predict 4FGL-DR4 separately under each spectral representation.

Table 2. Adopted hyperparameters of the classifiers.

Hyperparameter	Value	Explanation
TabNet		
n_d	64	Dimension of the decision layer at each step.
n_a	64	Dimension of the attentive transformer that generates feature masks.
n_steps	5	Number of sequential decision steps.
gamma	1.5	Relaxation factor that controls feature reuse across decision steps.
n_independent	2	Number of independent GLU blocks in each feature transformer.
n_shared	2	Number of shared GLU blocks across decision steps.
lambda_sparse	10^{-4}	Strength of the sparsity penalty imposed on the attention masks.
optimizer	Adam	Gradient-based optimizer used to update the network weights.
learning_rate	0.02	Initial optimizer step size.
scheduler	StepLR	Learning-rate scheduler used during training.
step_size	10	Number of epochs between scheduler updates.
gamma_scheduler	0.9	Multiplicative decay factor applied by the scheduler.
mask_type	sparsemax	Activation used to generate sparse feature-selection masks.
max_epochs	100	Maximum number of training epochs.
patience	20	Early-stopping patience in epochs.
batch_size	1024	Number of samples in each mini-batch.
virtual_batch_size	128	Ghost-batch size for batch normalization.
Random Forest		
n_estimators	300	Number of decision trees in the forest.
max_depth	15	Maximum depth of each tree.
min_samples_split	5	Minimum number of samples required to split an internal node.
min_samples_leaf	2	Minimum number of samples required in a leaf node.
class_weight	balanced	Class weighting inversely proportional to training-set frequencies.
Logistic-Regression Meta-classifier		
C	1.0	Inverse regularization strength that controls coefficient shrinkage.
solver	lbfgs	Numerical optimizer used to fit the multiclass logistic-regression model.
max_iter	100	Maximum number of iterations allowed for solver convergence.

3. MODEL ARCHITECTURE, HYPERPARAMETERS, AND PERFORMANCE METRICS

In this section, we describe the three classifiers used in this work: TabNet, Random Forest (RF), and a stacked ensemble that combines TabNet and RF. We also define the metrics used to evaluate model performance on the validation set. We adopted fixed hyperparameter settings for the final experiments. We used the same settings across the PL, LP, and PLEC representations to isolate performance differences associated with spectral representation rather than representation-specific hyperparameter retuning (S. O. Arik & T. Pfister 2021; L. Breiman 2001; F. Pedregosa et al. 2011). Tab. 2 summarizes the adopted hyperparameter settings for TabNet, RF, and the logistic-regression meta-classifier. In this table, Col. (1) lists the hyperparameter, Col. (2) gives the adopted value, and Col. (3) states its function. Group headers identify the corresponding model.

3.1. TabNet

TabNet targets tabular data and performs sequential attentive feature selection (S. O. Arik & T. Pfister 2021). Instead of assigning equal importance to all input features at every layer, TabNet learns sparse decision masks across multiple decision steps. This design allows the model to focus on the most informative dimensions of each source feature vector. It also aligns well with catalog-based classification, as a fixed set of tabular attributes describes each source. We train TabNet in a PyTorch ecosystem (A. Paszke et al. 2019).

Recent astrophysical studies have already explored TabNet for globular-cluster detection, brown-dwarf searches, and stellar-age estimation (D. Dold & K. Fahrion 2022; A. Avdeeva 2023; H. Zhang et al. 2024). Researchers still use TabNet less often in astrophysics than they use more established tree-based methods. We use TabNet here because its architecture matches the tabular structure of *Fermi*-LAT catalog features and because its application to fine-grained multiclass classification of unresolved γ -ray sources remains limited.

Let $\mathbf{x} \in \mathbb{R}^m$ denote the input feature vector. At decision step t , TabNet computes a feature mask $\mathbf{M}^{(t)} \in [0, 1]^m$ and forms the masked representation

$$\tilde{\mathbf{x}}^{(t)} = \mathbf{M}^{(t)} \odot \mathbf{x}, \quad (8)$$

where $\odot$ denotes element-wise multiplication. The feature transformer maps $\tilde{\mathbf{x}}^{(t)}$ to a latent decision representation $\mathbf{d}^{(t)}$. After t decision steps, the model aggregates the decision representations and maps them to class logits. For a C -class problem, TabNet outputs

$$\mathbf{p}_{\text{tab}}(\mathbf{x}) = \text{softmax} \left(\sum_{t=1}^T \mathbf{W}^{(t)} \mathbf{d}^{(t)} + \mathbf{b} \right), \quad (9)$$

and predicts

$$\hat{y}_{\text{tab}}(\mathbf{x}) = \arg \max_{c \in \{1, \dots, C\}} p_{\text{tab}, c}(\mathbf{x}). \quad (10)$$

In this work, we apply TabNet to each spectral representation introduced in Sec. 2.

3.2. Random Forest

RF combines a large set of decorrelated decision trees and aggregates their predictions through majority voting (L. Breiman 2001). For multiclass classification, if $\{h_t(\mathbf{x})\}_{t=1}^{T_{\text{rf}}}$ denotes the tree ensemble, then RF predicts

$$\hat{y}_{\text{rf}}(\mathbf{x}) = \arg \max_{c \in \{1, \dots, C\}} \sum_{t=1}^{T_{\text{rf}}} \mathbb{I}[h_t(\mathbf{x}) = c], \quad (11)$$

where $\mathbb{I}(\cdot)$ denotes the indicator function. RF estimates the class probability as

$$p_{\text{rf}, c}(\mathbf{x}) = \frac{1}{T_{\text{rf}}} \sum_{t=1}^{T_{\text{rf}}} \mathbb{I}[h_t(\mathbf{x}) = c]. \quad (12)$$

RF handles structured catalog data well because it captures nonlinear feature interactions, tolerates moderate noise, and remains competitive when feature distributions differ across classes. Astronomers have already applied RF to multiwavelength source classification, Galactic Center stellar classification, white dwarf classification from *Gaia* spectra, and photometric quasar classification (D. Gao et al. 2009; P. M. Plewa 2018; E. M. García-Zamora et al. 2023; B. Arroquia-Cuadros et al. 2023). These studies show that RF provides a strong baseline for tabular astrophysical data. We implement RF and the meta-classifier with scikit-learn (F. Pedregosa et al. 2011).

3.3. Stacked Ensemble Learning

We use stacked ensemble learning to combine the complementary strengths of TabNet and RF. TabNet supplies a flexible nonlinear model for tabular data, whereas RF provides a robust tree-based baseline. Previous astrophysical studies have shown that ensemble methods can improve performance in catalog-based classification and event reconstruction tasks. Researchers have used stacking or related ensemble strategies for star/galaxy classification, star/galaxy/quasar classification with photometric redshift assistance, and Cherenkov telescope event reconstruction (E. J. Kim et al. 2015; C. Li et al. 2020; P. A. C. Cunha & A. Humphrey 2022; A. Pagliaro et al. 2023). These results motivate our use of a stacked architecture in the present problem.

For each spectral representation, we split the training data into a base-model subset and a meta-model subset in a 7:3 ratio. We train TabNet and RF on the base-model subset, generate class-probability outputs for the meta-model subset, and use these outputs to train the logistic-regression meta-classifier. Within the base-model subset, we further reserve 10% of the data for TabNet validation. We keep the external validation set fully separate from all base-model and meta-model training steps and apply the trained stacked model to it only after completing all training steps. We do not retrain the base learners after fitting the meta-classifier. This design avoids fitting the meta-classifier on in-sample base-model predictions.

Let

$$\mathbf{p}_{\text{tab}}(\mathbf{x}) = [p_{\text{tab}, 1}(\mathbf{x}), \dots, p_{\text{tab}, C}(\mathbf{x})] \quad (13)$$

and

$$\mathbf{p}_{\text{rf}}(\mathbf{x}) = [p_{\text{rf}, 1}(\mathbf{x}), \dots, p_{\text{rf}, C}(\mathbf{x})] \quad (14)$$

denote the class-probability vectors produced by TabNet and RF. We concatenate them into the meta-feature vector

$$\mathbf{z}(\mathbf{x}) = [\mathbf{p}_{\text{tab}}(\mathbf{x}), \mathbf{p}_{\text{rf}}(\mathbf{x})] \in \mathbb{R}^{2C}. \quad (15)$$

The meta-model then computes

$$p_{\text{ens},c}(\mathbf{x}) = \frac{\exp(\mathbf{w}_c^\top \mathbf{z}(\mathbf{x}) + b_c)}{\sum_{k=1}^C \exp(\mathbf{w}_k^\top \mathbf{z}(\mathbf{x}) + b_k)}, \quad (16)$$

and predicts

$$\hat{y}_{\text{ens}}(\mathbf{x}) = \arg \max_{c \in \{1, \dots, C\}} p_{\text{ens},c}(\mathbf{x}). \quad (17)$$

3.4. Performance Metrics

We evaluate all models on the validation set with standard multiclass classification metrics. Because the class distribution is highly imbalanced, we report overall accuracy together with weighted Precision, Recall, and F_1 score. We compute these metrics with scikit-learn (F. Pedregosa et al. 2011).

Let N_{val} denote the number of validation samples, and let y_i and $\hat{y}_i$ denote the true and predicted labels of the i th validation source.

3.4.1. Accuracy

We define the overall accuracy as

$$\text{Accuracy} = \frac{1}{N_{\text{val}}} \sum_{i=1}^{N_{\text{val}}} \mathbb{I}(\hat{y}_i = y_i). \quad (18)$$

3.4.2. Precision, Recall, and F_1 Score

For class c , let TP_c , FP_c , and FN_c denote the true-positive, false-positive, and false-negative counts. We define

$$\text{Precision}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c}, \quad (19)$$

$$\text{Recall}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c}, \quad (20)$$

and

$$F1_c = \frac{2 \text{Precision}_c \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}. \quad (21)$$

3.4.3. Weighted Metrics

Because the validation set contains very unequal class supports, we compute weighted averages with the number of validation samples in each class, denoted by n_c :

$$\text{Precision}_w = \sum_{c=1}^C \frac{n_c}{N_{\text{val}}} \text{Precision}_c, \quad (22)$$

$$\text{Recall}_w = \sum_{c=1}^C \frac{n_c}{N_{\text{val}}} \text{Recall}_c, \quad (23)$$

$$F1_w = \sum_{c=1}^C \frac{n_c}{N_{\text{val}}} F1_c. \quad (24)$$

We define the macro average of a class-wise metric Q_c by

$$Q_{\text{macro}} = \frac{1}{C} \sum_{c=1}^C Q_c, \quad (25)$$

which assigns equal weight to each class.

These weighted and macro metrics provide a consistent basis for comparing TabNet, RF, and the ensemble model on imbalanced multiclass data.

3.4.4. Confusion Matrix

To inspect class-dependent prediction behavior, we first define the raw confusion matrix

$$\mathbf{M} \in \mathbb{N}^{C \times C}, \quad M_{ab} = \sum_{i=1}^{N_{\text{val}}} \mathbb{I}(y_i = a, \hat{y}_i = b), \quad (26)$$

where row a denotes the true class and column b denotes the predicted class. We then normalize each row to obtain the row-normalized confusion matrix

$$\tilde{M}_{ab} = \frac{M_{ab}}{\sum_{k=1}^C M_{ak}}, \quad (27)$$

so that each row gives the conditional distribution of predicted labels for a fixed true class. This normalization highlights class-dependent misclassification patterns and reduces the visual dominance of large classes.

3.4.5. ROC Curves and AUC

We also evaluate probabilistic discrimination through one-versus-rest ROC curves. For each class c , we convert the multiclass target into a binary label,

$$y_i^{(c)} = \begin{cases} 1, & y_i = c, \\ 0, & y_i \neq c, \end{cases} \quad (28)$$

and then compute the true-positive rate

$$\text{TPR}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FN}_c} \quad (29)$$

and false-positive rate

$$\text{FPR}_c = \frac{\text{FP}_c}{\text{FP}_c + \text{TN}_c}. \quad (30)$$

We define the area under the ROC curve (AUC) as

$$\text{AUC}_c = \int_0^1 \text{TPR}_c(\text{FPR}) d(\text{FPR}). \quad (31)$$

For the full multiclass task, we further report micro- and macro-averaged ROC-AUC values.

In Sec. 4, we use these metrics to compare model performance across the PL, LP, and PLEC representations.

3.4.6. Probability margins and predictive entropy

For unresolved-source predictions, we report the full ensemble probability vector rather than only the argmax class. Let $p_{(1)}$ and $p_{(2)}$ be the largest and second-largest predicted probabilities for a source. We define the top-1-top-2 probability margin as

$$\Delta p = p_{(1)} - p_{(2)}. \quad (32)$$

A large margin indicates that the most probable class is well separated from the second choice, whereas a small margin identifies a boundary source. We also compute the predictive entropy

$$H = - \sum_{c=1}^C p_{\text{ens},c} \ln p_{\text{ens},c}, \quad (33)$$

where $C = 22$. Low entropy indicates a concentrated probability distribution, while high entropy indicates that probability is spread across several candidate classes. These two quantities are especially useful for interpreting BCU subclassification and the BLL-FSRQ boundary.

4. EXPERIMENTAL PROCEDURE, VALIDATION RESULTS, AND PREDICTIONS

In this section, we describe the final experimental procedure and present the validation and prediction results for the PL, LP, and PLEC representations defined in Sec. 2. We follow the model framework introduced in Sec. 3 and apply the same processing logic to the three spectral representations.

4.1. Feature construction and preprocessing

For each spectral representation $s \in \{\text{LP}, \text{PL}, \text{PLEC}\}$, we denote the labeled sample by

$$\mathcal{D}^{(s)} = \left\{ (\mathbf{x}_i^{(s)}, y_i) \right\}_{i=1}^{N_s}, \quad (34)$$

where $\mathbf{x}_i^{(s)}$ denotes the feature vector of the i th labeled source and y_i denotes its class label. We denote the unknown-source sample by

$$\mathcal{U}^{(s)} = \left\{ \mathbf{u}_j^{(s)} \right\}_{j=1}^{M_s}. \quad (35)$$

We construct the numerical feature matrix directly from the processed 4FGL-DR4 tables. In the baseline PL, LP, and PLEC pipelines, we remove the source-name column, the class-label column, all string-type columns, all uncertainty or error columns, and the history-type columns that record auxiliary fitting or catalog-tracking information rather than intrinsic source properties. These operations reduce redundancy and keep the input space focused on physically interpretable observables. In the uncertainty-aware ablation experiments, we retain the relevant catalog uncertainty columns and treat them as measurement-quality descriptors. These columns are standardized in the same way as the physical features and are not interpreted as astrophysical class labels. We adopt the catalog quantities and their physical definitions from the 4FGL, 4FGL-DR3, and 4FGL-DR4 catalog papers (S. Abdollahi et al. 2020, 2022; J. Ballet et al. 2023).

We then split the integrated band quantities into scalar features. For each source, the band-integrated photon-flux sequence and the spectral-energy-distribution sequence become

$$\mathbf{F}_i = [F_{i,1}, F_{i,2}, \dots, F_{i,8}], \quad (36)$$

and

$$\mathbf{S}_i = [(\nu F_\nu)_{i,1}, (\nu F_\nu)_{i,2}, \dots, (\nu F_\nu)_{i,8}]. \quad (37)$$

Accordingly, we treat `Flux_Band1–Flux_Band8` and `nuFnu_Band1–nuFnu_Band8` as independent scalar features.

For spectral representation s , we write the final feature vector as

$$\mathbf{x}_i^{(s)} = [\mathbf{x}_{i,\text{common}}, \mathbf{x}_{i,\text{spec}}^{(s)}, \mathbf{F}_i, \mathbf{S}_i], \quad (38)$$

where $\mathbf{x}_{i,\text{common}}$ contains the common scalar catalog features and $\mathbf{x}_{i,\text{spec}}^{(s)}$ contains the spectral-model-specific parameters. Tab. 6 lists the adopted features.

We fill missing values with feature-wise medians. For feature f_j , we define

$$x'_{ij} = \begin{cases} x_{ij}, & x_{ij} \neq \text{NaN}, \\ \text{median}(\{x_{kj} : x_{kj} \neq \text{NaN}\}), & x_{ij} = \text{NaN}. \end{cases} \quad (39)$$

We then augment minority classes through a local interpolation scheme. For each class with fewer than a preset minimum number of samples, we generate synthetic sources until that class reaches the target count. In the final runs, we set the target minimum to 3000 samples per class. For a given minority-class source $\mathbf{x}$, we select a nearby source $\mathbf{x}_n$ in feature space and generate a synthetic sample as

$$\mathbf{x}_{\text{syn}} = \mathbf{x} + \boldsymbol{\lambda} \odot (\mathbf{x}_n - \mathbf{x}), \quad (40)$$

where each component of $\boldsymbol{\lambda}$ is drawn independently from a uniform distribution on $[0, 0.3]$, and $\odot$ denotes element-wise multiplication. This restricted interpolation range keeps each synthetic sample close to the original minority-class source while modestly expanding local support in feature space. We therefore treat this procedure as an algorithmic class-balancing device for tabular learning rather than as a physically generative model of LAT source populations (L. Breiman 2001; S. O. Arik & T. Pfister 2021; A. Bhat & D. Malyshev 2022; D. Malyshev & A. Bhat 2023).

The baseline pipelines use the upper bound $\lambda_{\text{max}} = 0.3$. To evaluate whether this choice affects the conclusions, we also scan $\lambda_{\text{max}} = 0.1, 0.2, 0.3, 0.4$, and 0.5 in the feature-set and uncertainty-ablation experiments. The best value is then reported separately for each feature configuration rather than assumed to be universal.

After augmentation, each spectral representation expands from 3009 labeled sources to 66,000 samples. We then divide the augmented sample into training and validation subsets such that all synthetic samples remain in the training set and only real sources enter the validation set. In particular, for classes with a single real labeled source, that source is assigned to the validation set, and the corresponding synthetic samples remain in the training set. Because this procedure uses a narrow interpolation range, we treat it as a local data-expansion strategy rather than as a physically motivated generative model of source populations, and we therefore interpret rare-class results with appropriate caution (A. Bhat & D. Malyshev 2022; D. Malyshev & A. Bhat 2023; J. Coronado-Blázquez 2022). Let $\mathcal{R}_c^{(s)}$ and $\mathcal{S}_c^{(s)}$ denote the real and synthetic samples of class c . We write

$$\mathcal{D}_{\text{train}}^{(s)} = \bigcup_{c=1}^C \left(\mathcal{R}_{c,\text{train}}^{(s)} \cup \mathcal{S}_c^{(s)} \right), \quad (41)$$

and

$$\mathcal{D}_{\text{val}}^{(s)} = \bigcup_{c=1}^C \mathcal{R}_{c,\text{val}}^{(s)}, \quad (42)$$

where C denotes the number of target classes.

Finally, we standardize each feature with the statistics of the training set,

$$z_{ij} = \frac{x'_{ij} - \mu_j}{\sigma_j}, \quad (43)$$

where

$$\mu_j = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} x'_{ij}, \quad (44)$$

and

$$\sigma_j = \sqrt{\frac{1}{N_{\text{train}} - 1} \sum_{i=1}^{N_{\text{train}}} (x'_{ij} - \mu_j)^2}. \quad (45)$$

4.2. Feature ranking

We rank features with a Random Forest model and retain those whose importance exceeds the threshold $\tau = 0.01$ (L. Breiman 2001). This threshold removes low-contribution features while preserving the compact feature set that consistently supports strong validation performance across the three representations. Feature ranking is performed independently for each spectral representation using only the training split after imputation, augmentation, splitting, and standardization; we never use the validation or prediction samples in this step. If I_j denotes the importance of feature f_j , then the selected feature set is

$$\mathcal{F}_{\text{sel}} = \{f_j : I_j > \tau\}. \quad (46)$$

Tab. 7 lists the 10 highest-ranked features for the uncertainty-aware PL, LP, and PLEC configurations. In this table, Col. (1) gives the rank ordered by Random-Forest feature importance, Col. (2) gives the feature names at the same rank for the PL/LP/PLEC configurations, and Col. (3) gives the corresponding normalized importance values in the same PL/LP/PLEC order. The rankings show that the improved uncertainty-aware models are not driven by a single feature family. Pivot energy, variability, spectral-shape parameters, the one-hot encoded catalog spectral type, and uncertainty columns all appear among the high-ranked features. Tab. 8 summarizes representative ranking patterns across selected baseline and ablation configurations. In Table 8, Col. (1) identifies the experimental configuration, Col. (2) lists representative high-ranked features in that configuration, and Col. (3) summarizes the feature family or physical information represented by those features. In the no-uncertainty AllSpec configuration, `Pivot_Energy`, `Frac_Variability`, `PLEC_ExpfactorS`, `LP_Index`, `PL_Index`, and `LP_beta` all rank highly, showing that the classifier uses information from multiple spectral parameterizations together with variability. In the uncertainty-aware PLEC configuration, `Pivot_Energy`, `PLEC_ExpfactorS`, `Unc_PLEC_Flux_Density`, `Frac_Variability`, `SpectrumType_PLSuperExpCutoff`, `PLEC_IndexS`, and `Unc_PLEC_EPeak` are highly ranked. The photometry-plus-variability configuration is dominated by `Variability_Index`, `Frac_Variability`, band SED points, and band fluxes. Thus the classifier is not driven exclusively by either spectral shape or variability; spectral parameters, temporal behavior, broadband flux information, catalog spectral type, and measurement-quality descriptors jointly shape the final decision boundary.

Table 3. Best validation performance with and without uncertainty columns in the feature-set ablation experiments.

Feature set	Without uncertainty				With uncertainty				$\Delta F_{1,w}$
	$\lambda_{\max}$	Acc.	$F_{1,w}$	Log loss	$\lambda_{\max}$	Acc.	$F_{1,w}$	Log loss	
AllSpec	0.1	0.9430	0.9434	0.1376	0.1	0.9581	0.9582	0.1189	+0.0148
LP-only	0.1	0.9513	0.9516	0.1433	0.1	0.9564	0.9568	0.1189	+0.0052
PL-only	0.1	0.9446	0.9449	0.1516	0.2	0.9480	0.9486	0.1490	+0.0037
PLEC-only	0.1	0.9530	0.9531	0.1370	0.2	0.9614	0.9614	0.1309	+0.0084
P+V	0.1	0.9547	0.9548	0.1223	0.3	0.9530	0.9536	0.1505	-0.0013

The last column reports the change in weighted F_1 after adding uncertainty columns. Positive values indicate improvement relative to the corresponding uncertainty-free configuration.

Table 4. Weighted F_1 sensitivity to $\lambda_{\max}$ for uncertainty-aware feature configurations.

$\lambda_{\max}$	AllSpec	LP-only	PL-only	PLEC-only	P+V
0.1	0.9582	0.9568	0.9452	0.9517	0.9514
0.2	0.9403	0.9386	0.9486	0.9614	0.9434
0.3	0.9428	0.9502	0.9353	0.9359	0.9536
0.4	0.9316	0.9422	0.9323	0.9342	0.9329
0.5	0.9276	0.9306	0.9292	0.9214	0.9318

Only uncertainty-aware configurations are shown here because the preceding comparison identifies them as the main expanded models for the spectral-parameter configurations.

4.3. Validation metrics

The expanded feature-set and uncertainty-ablation experiments provide the main robustness test of the classification design. Table 3 summarizes the best validation results for AllSpec, LP-only, PL-only, PLEC-only, and photometry-plus-variability configurations, each tested with and without uncertainty columns. We use the abbreviation P+V for the photometry-plus-variability configuration in the tables; this configuration uses the non-spectral photometric and variability quantities without model-specific PL, LP, or PLEC spectral-shape parameters. In Table 3, Col. (1) names the feature configuration, Cols. (2)–(5) give the best scanned value of $\lambda_{\max}$, validation accuracy, weighted F_1 score, and multiclass log loss for the uncertainty-free version, Cols. (6)–(9) give the same four quantities for the uncertainty-aware version, and Col. (10) gives the change in weighted F_1 between the two versions. The log loss is the multiclass cross-entropy computed from the predicted class-probability vector and the true validation label, so lower values indicate better calibrated probability assignments. The strongest validation performance in the expanded analysis is obtained by the PLEC-only uncertainty-aware configuration at $\lambda_{\max} = 0.2$, with accuracy and weighted F_1 both equal to 0.9614. The AllSpec uncertainty-aware configuration is also competitive, with accuracy 0.9581 and weighted $F_1 = 0.9582$. Relative to the uncertainty-free configurations, adding uncertainty columns improves the weighted F_1 score for AllSpec, LP-only, PL-only, and PLEC-only. The photometry-plus-variability baseline remains strong, with weighted $F_1 = 0.9536$ when uncertainty columns are included. Thus, temporal and broadband flux information alone carry substantial class information, while spectral-shape and uncertainty/quality descriptors further improve or stabilize the spectral-parameter-based models. Figure 7 visualizes the with-/without-uncertainty comparison.

Table 4 gives the weighted- F_1 sensitivity to $\lambda_{\max}$ for the uncertainty-aware configurations. In this table, Col. (1) lists the upper bound of the synthetic-interpolation coefficient, and Cols. (2)–(6) report the weighted F_1 values obtained by the AllSpec, LP-only, PL-only, PLEC-only, and P+V uncertainty-aware configurations, respectively. The best value is not identical across feature sets: AllSpec and LP-only prefer $\lambda_{\max} = 0.1$, PL-only and PLEC-only prefer $\lambda_{\max} = 0.2$,

and photometry-plus-variability prefers $\lambda_{\max} = 0.3$. This test shows that the interpolation strength should be treated as an empirical balancing parameter rather than fixed without evaluation. Figure 8 shows the corresponding trend.

The three uncertainty-aware single spectral representations have the same labeled-sample and validation sizes. Each representation starts from 3009 labeled sources, expands to 66,000 samples after augmentation, uses 65,404 samples for training, validates on 596 real sources, and predicts 4186 unresolved sources. Tab. 9 summarizes the validation performance of the uncertainty-aware PL, LP, and PLEC ensemble pipelines at their best scanned $\lambda_{\max}$ values. In this table, Col. (1) identifies the adopted spectral parameterization, Col. (2) gives the selected $\lambda_{\max}$ value, Col. (3) gives the number of selected input variables, Col. (4) gives the number of training samples after augmentation, Col. (5) gives the number of real validation sources, Col. (6) gives the number of unresolved sources to be classified, and Cols. (7)–(11) report the aggregate validation metrics defined in Sec. 3.4. The uncertainty-aware PLEC configuration yields the strongest single-representation performance, with accuracy and weighted F_1 both equal to 0.9614, followed by LP and PL. These results motivate the use of the PLEC uncertainty-aware model as the final probabilistic classifier, while retaining AllSpec and the other configurations as controlled comparisons.

We caution against over-interpreting the class-wise scores of minority classes. Tab. 10 lists the validation support of each class explicitly for the final PLEC uncertainty-aware model. Several rare classes contribute only one or a few validation sources, so perfect or near-perfect precision, recall, and F_1 values in those rows do not imply the same level of statistical robustness as the corresponding scores of the dominant classes. The *gc* row is an extreme case: the class remains in the supervised target space, but it has only one real labeled source in the validation set. It therefore cannot support a meaningful estimate of class-specific performance. The class-wise results are therefore most informative for the dominant populations, especially *b11*, *fsrq*, *psr*, and *msh*. The very high macro F_1 value partly reflects the equal weighting of low-support classes in the macro average, including several classes that achieve perfect scores on very small validation counts. We therefore place greater interpretive weight on weighted metrics and on the major classes than on macro-averaged scores alone, following the general caution in imbalanced multiclass classification that aggregate metrics can obscure the statistical fragility of very low-support classes (F. Pedregosa et al. 2011; A. Bhat & D. Malyshev 2022; D. Malyshev & A. Bhat 2023).

Tab. 10 gives the class-wise precision, recall, and F_1 score of the final PLEC uncertainty-aware model. In this table, Col. (1) gives the class label, Col. (2) gives the validation support of that class, and Cols. (3)–(5) give precision, recall, and F_1 , respectively. The dominant classes, especially *b11* and *fsrq*, account for most of the global error budget, yet the classifier maintains stable performance on both. The final PLEC uncertainty-aware model reaches $(P, R, F_1) = (0.97, 0.96, 0.96)$ for *b11* and $(0.93, 0.93, 0.93)$ for *fsrq*. The model also performs strongly on *msh* and *psr*, with F_1 scores of 0.99 and 1.00, respectively.

4.4. Row-normalized confusion matrices

To visualize class-dependent prediction behavior, we use the row-normalized confusion matrix

$$\tilde{M}_{ab} = \frac{M_{ab}}{\sum_{k=1}^C M_{ak}}, \quad (47)$$

where row a denotes the true class and column b denotes the predicted class. Each row, therefore, gives the conditional distribution of predicted labels for a fixed true class.

Figs. 1–3 in the appendix show the row-normalized confusion matrices for the PL, LP, and PLEC representations. All three matrices remain strongly diagonal, which indicates that the ensemble captures the dominant class structure of the catalog. The largest off-diagonal terms appear in the *BLL*–*FSRQ* block. In LP, 89.9% of true *b11* sources remain on the diagonal and 6.4% move into *fsrq*, while 91.5% of true *fsrq* sources remain on the diagonal and 7.3% move into *b11*. PL shows a very similar pattern. PLEC reduces this overlap slightly, with 91.9% and 93.3% diagonal fractions for *b11* and *fsrq*, respectively. Outside the blazar block, the remaining confusion stays weak and local. These matrices therefore confirm that the main difficulty in classification still stems from the physical overlap between the two dominant blazar subclasses.

4.5. ROC–AUC curves

Figs. 4–6 in the appendix show the one-versus-rest ROC curves for the three spectral representations. The class-dependent curves cluster near the upper-left corner in all three cases. The micro- and macro-averaged AUC values also remain close to unity. These curves confirm the class-wise results and show that the PLEC representation yields

Table 5. Example probabilistic predictions for unresolved sources from the best PLEC uncertainty-aware model.

Source	Original	Top-1	p_1	Top-2	p_2	Top-3	p_3	$p_1 - p_2$
4FGL J0000.3-7355	blank	bll	0.843	fsrq	0.140	rdg	0.003	0.703
4FGL J0000.5+0743	blank	fsrq	0.913	bll	0.079	sey	0.001	0.833
4FGL J0000.7+2530	blank	bll	0.961	fsrq	0.029	gal	0.001	0.932
4FGL J0001.2+4741	bcu	bll	0.955	fsrq	0.040	rdg	0.001	0.915
4FGL J0001.6+3503	blank	bll	0.723	fsrq	0.240	glc	0.008	0.483
4FGL J0002.1+6721c	unk	glc	0.272	mzp	0.253	spp	0.151	0.020

The full machine-readable table contains 4186 unresolved sources and the complete 22-class probability vector for each source.

the strongest overall discrimination among the three baseline models. The one-versus-rest ROC curves for the best uncertainty-aware PLEC model are shown in Fig. 10.

4.6. Predictions for unknown sources

After validation, we apply the trained ensemble classifier to the unknown-source sample $\mathcal{U}^{(s)}$ in each spectral representation. For source $\mathbf{u}_j^{(s)}$, the classifier outputs

$$\hat{y}(\mathbf{u}_j^{(s)}) = \arg \max_{c \in \{1, \dots, C\}} p_{\text{ens}, c}(\mathbf{u}_j^{(s)}), \quad (48)$$

where $\hat{y}(\mathbf{u}_j^{(s)})$ denotes the predicted subclass label.

Tab. 11 lists paired example top-1 subclass predictions for 20 unresolved sources under uncertainty-free and uncertainty-aware versions of the main feature configurations. In this table, Col. (1) gives the 4FGL-DR4 source name, Col. (2) gives the catalog unresolved-source flag before classification, and Cols. (3)–(7) give the paired predictions for PL, LP, PLEC, AllSpec, and P+V, respectively. Here P+V has the same meaning as above: the photometry-plus-variability configuration without model-specific spectral-shape parameters. Within each entry in Cols. (3)–(7), the class label before the slash is the uncertainty-free prediction and the class label after the slash is the uncertainty-aware prediction. The PLEC column therefore compares the baseline PLEC result with the final selected uncertainty-aware PLEC model, whereas the AllSpec column compares the simultaneous use of PL, LP, and PLEC parameters without and with uncertainty information. This format makes it possible to identify sources whose predicted labels are stable under the inclusion of uncertainty columns and sources whose classifications are sensitive to measurement-quality information. Stable labels indicate cases whose catalog observables consistently place them in a well-separated region of feature space, whereas changes across the slash identify boundary or low-information cases that should be interpreted with greater caution.

For the best PLEC uncertainty-aware model, the machine-readable prediction table provides a probabilistic classification for all 4186 unresolved sources. The most common top-1 predictions are **bll** (1558 sources), **fsrq** (1147), **spp** (778), **glc** (228), and **mzp** (201). Among the 1623 BCUs, 820 are predicted as **bll** and 597 as **fsrq**. The average and median top-1 probabilities are 0.832 and 0.903, respectively. However, 301 sources have top-1 probability below 0.5, and 332 sources have top-1–top-2 probability margin below 0.2. These sources are natural low-confidence or boundary candidates. Table 5 gives representative entries. In this table, Col. (1) gives the 4FGL-DR4 source name, Col. (2) gives the catalog unresolved-source flag, Cols. (3), (5), and (7) give the three most probable predicted classes, Cols. (4), (6), and (8) give their corresponding probabilities, and Col. (9) gives the top-1–top-2 probability margin.

Overall, the expanded with-/without-uncertainty comparison establishes that measurement-quality information improves the spectral-parameter-based configurations, and the uncertainty-aware PL, LP, and PLEC results identify the PLEC configuration as the strongest final model for the probabilistic unresolved-source catalog. The uncertainty-free results are therefore retained as controlled baselines, while the uncertainty-aware PLEC model is used for the final probability catalog.

Table 6. Definitions of the adopted spectral and non-spectral 4FGL-DR4 features.

Feature	Unit	Description
Pivot_Energy	GeV	Energy at which the differential-flux uncertainty reaches its minimum
Flux1000	$\text{cm}^{-2} \text{s}^{-1}$	Integrated photon flux above 1 GeV from spectral fitting
Energy_Flux100	$\text{erg cm}^{-2} \text{s}^{-1}$	Energy flux integrated over the catalog reference band
Variability_Index	...	Likelihood-based variability statistic over catalog time bins
Frac_Variability	...	Fractional excess variability beyond statistical and systematic fluctuations
Flux_Peak	$\text{cm}^{-2} \text{s}^{-1}$	Peak flux over the catalog variability intervals
Flux_Band1	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 50–100 MeV
Flux_Band2	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 100–300 MeV
Flux_Band3	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 300–1000 MeV
Flux_Band4	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 1–3 GeV
Flux_Band5	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 3–10 GeV
Flux_Band6	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 10–30 GeV
Flux_Band7	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 30–100 GeV
Flux_Band8	$\text{cm}^{-2} \text{s}^{-1}$	Photon flux in 100–1000 GeV
nuFnu_Band1	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 50–100 MeV
nuFnu_Band2	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 100–300 MeV
nuFnu_Band3	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 300–1000 MeV
nuFnu_Band4	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 1–3 GeV
nuFnu_Band5	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 3–10 GeV
nuFnu_Band6	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 10–30 GeV
nuFnu_Band7	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 30–100 GeV
nuFnu_Band8	$\text{erg cm}^{-2} \text{s}^{-1}$	Spectral energy distribution in 100–1000 GeV
LP_Flux_Density	$\text{cm}^{-2} \text{MeV}^{-1} \text{s}^{-1}$	Differential flux at pivot energy in the LogParabola model
LP_Index	...	Photon index at pivot energy in the LogParabola model
LP_beta	...	Curvature parameter in the LogParabola model
LP_EPeak	GeV	Peak energy of the νF_ν spectrum in the LogParabola model
PL_Flux_Density	$\text{cm}^{-2} \text{MeV}^{-1} \text{s}^{-1}$	Differential flux at pivot energy in the PowerLaw model
PL_Index	...	Photon index in the PowerLaw model
PLEC_Flux_Density	$\text{cm}^{-2} \text{MeV}^{-1} \text{s}^{-1}$	Differential flux at pivot energy in the PLSuperExpCutoff model
PLEC_IndexS	...	Local photon index at pivot energy in the PLSuperExpCutoff model
PLEC_ExpfactorS	...	Local curvature or cutoff-shape parameter in the PLSuperExpCutoff model
PLEC_EPeak	GeV	Peak energy of the νF_ν spectrum in the PLSuperExpCutoff model

Common features apply to all three representations; PL, LP, and PLEC each add their model-specific parameters. LP and PLEC each retain 26 features, whereas PL retains 24 features.

Table 6 defines the spectral and non-spectral 4FGL-DR4 features used in the processed tables. In this table, Col. (1) gives the exact processed column name, Col. (2) gives the physical unit or indicates that the quantity is dimensionless, and Col. (3) gives the operational catalog meaning used for model input construction.

5. DISCUSSION AND CONCLUSIONS

5.1. Discussion

Our results show that spectral representation changes multiclass classification outcomes in ways consistent with the known spectral behavior of major LAT source populations. The three pipelines use the same source set and learning framework, but they yield different performance levels and different subclass assignments. The PL, LP, and PLEC parameterizations encode different aspects of the γ -ray spectrum and therefore define different feature spaces. Among the three representations, PLEC delivers the strongest overall performance, with the highest accuracy and weighted F_1 score in Sec. 4. This result indicates that the local-slope and cutoff-sensitive form of PLEC separates the dominant classes more effectively than either PL or LP (S. Abdollahi et al. 2022; J. Ballet et al. 2023).

The class-wise results identify the main astrophysical difficulty of the problem. In all three spectral representations, the classifier separates most non-blazar classes well, but it still shows the largest confusion between BLL and FSRQ. We expect this outcome on physical grounds. Both classes belong to the blazar population, both are jet-dominated, and both occupy overlapping regions in LAT parameter space. The row-normalized confusion matrices in Sec. 4 and Figs. 1–3 confirm this point: the largest off-diagonal terms lie in the BLL–FSRQ block, while the remaining confusion

Table 7. Top 10 ranked features in the uncertainty-aware PL, LP, and PLEC configurations.

Rank	Feature (PL/LP/PLEC)	Importance (PL/LP/PLEC)
1	Pivot_Energy/Pivot_Energy/Pivot_Energy	0.045/0.036/0.041
2	Unc_PL_Flux_Density/LP_beta/PLEC_ExpfactorS	0.038/0.032/0.037
3	Frac_Variability/Frac_Variability/Unc_PLEC_Flux_Density	0.034/0.031/0.034
4	SpectrumType_PLSuperExpCutoff/SpectrumType_PLSuperExpCutoff/Frac_Variability	0.033/0.029/0.031
5	PL_Index/Unc_LP_Flux_Density/SpectrumType_PLSuperExpCutoff	0.033/0.028/0.030
6	Variability_Index/LP_EPeak/PLEC_IndexS	0.032/0.027/0.029
7	Unc_Flux_Band1_pos/LP_SigCurv/Unc_PLEC_EPeak	0.029/0.027/0.028
8	Unc_Flux_Band3_pos/LP_Index/Unc_Flux_Band8_pos	0.028/0.026/0.026
9	Unc_Flux_Band8_pos/Unc_Flux_Band3_pos/PLEC_EPeak	0.027/0.026/0.026
10	Unc_Flux_Band4_neg/Unc_Flux_Band1_pos/PLEC_Flux_Density	0.027/0.026/0.026

The ranking is recomputed separately for each uncertainty-aware single-representation configuration at the best scanned $\lambda_{\max}$ value for that configuration.

Table 8. Representative feature-ranking patterns.

Configuration	Representative high-ranked features	Interpretation
PL baseline, no uncertainty	Variability_Index; Frac_Variability nuFnu_Band8; Flux_Band8 Pivot_Energy Variability_Index; LP_beta	Variability, high-energy flux, and PL information
LP baseline, no uncertainty	Frac_Variability; LP_Index Pivot_Energy Variability_Index; PLEC_ExpfactorS	Variability plus LP curvature and slope information
PLEC baseline, no uncertainty	Frac_Variability; nuFnu_Band8 Pivot_Energy Pivot_Energy; Frac_Variability PLEC_ExpfactorS; LP_Index	Variability plus cutoff-sensitive PLEC information
AllSpec, no uncertainty	PL_Index; LP_beta Pivot_Energy; PLEC_ExpfactorS Unc_PLEC_Flux_Density	Joint use of multiple spectral parameterizations
PLEC, with uncertainty	Frac_Variability; SpectrumType_PLEC PLEC_IndexS Variability_Index; Frac_Variability nuFnu_Band8; Flux_Band1	Spectral shape plus measurement quality
Photometry+Variability, with uncertainty	Unc_Flux_Band4_pos	Non-spectral baseline with band-flux quality descriptors

This table condenses the complete feature-importance CSV files into a manuscript-readable summary. The shorthand `SpectrumType_PLEC` denotes the one-hot encoded `SpectrumType_PLSuperExpCutoff` column.

Table 9. Overall validation performance of the uncertainty-aware ensemble classifier in the three single spectral representations.

Representation	$\lambda_{\max}$	Features	N_{train}	N_{val}	M_s	Accuracy	Precision _w	Recall _w	F1 _w	F1 _{macro}
PL	0.2	47	65404	596	4186	0.9480	0.9509	0.9480	0.9486	0.9758
LP	0.1	48	65404	596	4186	0.9564	0.9582	0.9564	0.9568	0.9795
PLEC	0.2	50	65404	596	4186	0.9614	0.9616	0.9614	0.9614	0.9924

The listed rows use the best scanned uncertainty-aware configuration for each single spectral representation.

537 stays weaker and more localized. The classifier therefore recovers the broad catalog-level class structure of the LAT
538 population, but the two dominant blazar classes still define the hardest class boundary (S. Abdollahi et al. 2020, 2022;
539 J. Ballet et al. 2023; M. Ajello et al. 2020, 2022).

Table 10. Class-wise validation metrics of the final PLEC uncertainty-aware model.

Class	Support	Precision	Recall	F1
agn	1	1.00	1.00	1.00
bin	2	1.00	1.00	1.00
bll	298	0.97	0.96	0.96
css	1	1.00	1.00	1.00
fsrq	164	0.93	0.93	0.93
gal	1	1.00	1.00	1.00
gc	1	1.00	1.00	1.00
glc	8	1.00	1.00	1.00
hmb	2	1.00	1.00	1.00
lmb	1	1.00	1.00	1.00
msp	35	0.97	1.00	0.99
nlsy1	1	1.00	1.00	1.00
nov	1	1.00	1.00	1.00
psr	28	1.00	1.00	1.00
pwn	4	1.00	1.00	1.00
rdg	10	0.91	1.00	0.95
sbg	1	1.00	1.00	1.00
sey	1	1.00	1.00	1.00
sfr	1	1.00	1.00	1.00
snr	8	1.00	1.00	1.00
spp	26	1.00	1.00	1.00
ssrq	1	1.00	1.00	1.00
Macro Avg	596	0.99	1.00	0.99
Weighted Avg	596	0.96	0.96	0.96
Accuracy	596		0.9614	

Macro Avg, Weighted Avg, and Accuracy are aggregate summary rows rather than physical source classes. Rare-class rows are retained for catalog completeness.

The pulsar-related classes show a different pattern. The model separates **psr** and **msp** more cleanly than it separates the blazar classes, especially under the PLEC representation. This result is consistent with the established LAT view that pulsar spectra commonly exhibit strong curvature or cutoff-like behavior (A. A. Abdo et al. 2013; S. Abdollahi et al. 2022; J. Ballet et al. 2023). In this sense, spectral representation serves not only as a computational choice but also as a physically meaningful description of differences among major source populations.

The feature-ranking results support the same interpretation. Variability-related quantities rank among the most important features in all three spectral representations, while model-specific spectral parameters also enter the top ranks. This combination matches the basic structure of the LAT sky. Variability separates strongly variable AGN-like populations from steadier Galactic emitters, while curvature and cutoff descriptors separate pulsar-like sources from power-law-dominated or smoothly curved extragalactic sources. The classifier does not rely on any single observable. Instead, it combines temporal and spectral information to recover class structure across a heterogeneous population of γ -ray sources (S. Abdollahi et al. 2020, 2022; L. Breiman 2001; S. O. Arik & T. Pfister 2021; M. Ackermann et al. 2013; S. Abdollahi et al. 2017).

Our definition of the unknown-source population sharpens the astrophysical question. We combine blank-class sources, **unk** sources, and BCUs into a single unresolved-source framework while preserving the distinct meaning of the BCU label. A BCU does not represent a fully unidentified source; it marks a source that remains unresolved mainly at the blazar-subclass level. By treating BCUs as subclass-unknown sources, we ask which catalog subclass each unresolved source most likely belongs to within a fine-grained taxonomy. This formulation more directly supports follow-up work because it yields direct subclass hypotheses rather than only broad source-family assignments (S. Abdollahi et al. 2022; J. Ballet et al. 2023; M. Ajello et al. 2020, 2022).

The source-level comparison in Sec. 4 shows that some unresolved sources retain the same top-1 label when uncertainty columns are added, whereas others change their predicted labels in one or more feature configurations. Stable

Table 11. Example paired subclass predictions for 20 unresolved sources.

Source	Original	PL	LP	PLEC	AllSpec	P+V
4FGL J0000.3-7355	blank	fsrq/bll	bll/bll	bll/bll	bll/bll	fsrq/bll
4FGL J0000.5+0743	blank	fsrq/fsrq	fsrq/fsrq	fsrq/fsrq	bll/bll	fsrq/fsrq
4FGL J0000.7+2530	blank	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0001.2+4741	bcu	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0001.6+3503	blank	bll/bll	bll/bll	bll/bll	glc/glc	fsrq/msp
4FGL J0001.6-4156	bcu	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0001.8-2153	bcu	bll/bll	bll/bll	fsrq/bll	bll/bll	bll/bll
4FGL J0002.1+6721c	unk	fsrq/glc	msp/spp	glc/glc	spp/spp	msp/msp
4FGL J0002.3-0815	bcu	bll/fsrq	bll/bll	bll/bll	bll/bll	spp/fsrq
4FGL J0002.4-5156	bcu	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0003.1-5248	bcu	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0003.3-0703	bcu	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0003.3+2511	blank	bll/rdg	fsrq/fsrq	fsrq/msp	bll/msp	msp/bll
4FGL J0003.3-1928	bcu	fsrq/fsrq	fsrq/fsrq	fsrq/fsrq	fsrq/fsrq	fsrq/fsrq
4FGL J0003.3-5905	bcu	fsrq/bll	fsrq/bll	bll/fsrq	bll/fsrq	bll/fsrq
4FGL J0003.5+0717	bcu	bll/bll	bll/bll	bll/bll	glc/bll	bll/bll
4FGL J0003.6+3059	blank	bll/msp	glc/msp	msp/msp	msp/msp	msp/msp
4FGL J0004.0+5715	blank	msp/bll	glc/fsrq	msp/fsrq	fsrq/spp	bll/msp
4FGL J0004.4-4001	blank	bll/bll	bll/bll	bll/bll	bll/bll	bll/bll
4FGL J0005.5+6634	blank	spp/spp	spp/spp	msp/spp	spp/spp	psr/psr

The complete machine-readable version contains all 4186 unresolved sources and also provides the top-1 probabilities, probability margins, and predictive entropies for the paired configurations.

paired predictions likely reflect sources whose catalog observables place them in relatively well-separated regions of feature space. Changes between the uncertainty-free and uncertainty-aware labels may indicate boundary objects, low-information sources, or sources whose LAT catalog measurements are sufficiently uncertain that measurement-quality descriptors alter the learned decision boundary. These cases deserve special attention in future follow-up work because they may trace unusual blazars, mixed populations, or sources whose LAT observables alone do not support a unique subclass assignment. This issue also appears in recent probabilistic and multiclass studies of LAT source classification (A. Bhat & D. Malyshev 2022; D. V. Malyshev 2023; J. Coronado-Blázquez 2022).

The ensemble comparison also yields a clear methodological result. The stacked ensemble outperforms either base learner alone, indicating that TabNet and RF capture complementary structures within the same tabular feature space. RF provides a strong baseline because it handles nonlinear feature interactions and moderate noise robustly, while TabNet learns sequential attention over tabular inputs. The logistic-regression meta-learner then combines the two probability outputs, thereby reducing model-specific bias. In this sense, the ensemble does not simply average two classifiers; it integrates two distinct views of the same data (L. Breiman 2001; S. O. Arik & T. Pfister 2021).

Our results fit naturally into the literature on statistical classification of LAT sources, but they fill a different niche. This broader literature also includes deep-learning and optimized neural-network approaches to LAT-source and BCU classification, which further support the view that the unresolved γ -ray sky contains recoverable structure in catalog observables (G. Chiaro et al. 2019; T. Finke et al. 2021; S. Germani et al. 2021). Early work already showed that statistical learning can classify unresolved γ -ray sources using catalog features, including Gaussian-mixture approaches in pulsar astronomy and Random Forest classification of unassociated LAT sources (K. J. Lee et al. 2012; N. Mirabal et al. 2012). More recent studies extended this line to probabilistic and multiclass frameworks (A. Bhat & D. Malyshev 2022; S. Germani et al. 2021; D. Malyshev & A. Bhat 2023; D. V. Malyshev 2023). Within that line of work, our study contributes a fine-grained 4FGL-DR4 analysis in which spectral representation itself becomes part of the classification design.

D. Malyshev & A. Bhat (2023) developed a hierarchical multiclass framework on 4FGL-DR3 and grouped the physical classes into coarser structures, such as 6-class and 9-class systems, in order to stabilize classification when some original classes contain very few labeled sources. Our work takes a different path. We maintain a direct, fine-grained catalog

taxonomy, work on 4FGL-DR4, and evaluate how spectral representation itself affects classification outcomes. We therefore view the two studies as complementary rather than directly competitive.

J. Coronado-Blázquez (2022) provides a second useful comparison because that work also attempted direct catalog-level fine-grained classification on 4FGL-DR3. In that sense, it lies closer to our task definition than the hierarchical framework of D. Malyshev & A. Bhat (2023). Even so, the comparison still has limits. J. Coronado-Blázquez (2022) worked on 4FGL-DR3, used CatBoost, and reported global performance for a 23-class setup. In contrast, we work on 4FGL-DR4, use a stacked TabNet+RF framework, and study how PL, LP, and PLEC representations change the final subclass assignments. We therefore use that work as a methodological and task-level reference rather than as a source-by-source benchmark.

The expanded analysis clarifies the role of spectral information without discarding the original PL, LP, and PLEC framework. The baseline single-representation pipelines show that each catalog spectral form contains useful class information, with PLEC giving the strongest baseline performance. The AllSpec result confirms that using all PL, LP, and PLEC parameters simultaneously is a valid and competitive strategy. At the same time, the uncertainty-aware PLEC configuration produces the best validation score, suggesting that cutoff-sensitive parameters are particularly informative for this catalog-level task. The photometry-plus-variability baseline remains strong, which shows that temporal and broadband flux information carry substantial class information and prevents an overinterpretation of the classifier as relying only on model-specific spectral parameters.

The feature-importance comparison also resolves the apparent tension between the importance of variability and the use of spectral parameters. In the baseline PL, LP, and PLEC rankings, variability-related quantities rank highly, confirming the discriminating power of temporal information. In the expanded runs, however, AllSpec and uncertainty-aware PLEC also assign high importance to spectral-shape parameters and uncertainty/quality descriptors. The classification is therefore not driven by a single feature family. Variability and broadband photometry provide a strong base, while spectral shape, catalog spectral type, and measurement-quality information add complementary information to the decision boundary.

The uncertainty comparison directly shows that measurement-quality information affects classification performance. Adding uncertainty columns improves the best weighted F_1 score for AllSpec, LP-only, PL-only, and PLEC-only configurations. These uncertainty columns are catalog-level quality descriptors; they do not replace a full propagation of spectral-fit posterior distributions, but they provide an empirical test of whether ignoring measurement uncertainty changes the learned decision boundary. The probabilistic output further changes how the unresolved-source catalog should be used. A source assigned to `b11` with $p_1 = 0.95$ and $p_1 - p_2 = 0.90$ is qualitatively different from a source assigned to `b11` with $p_1 = 0.52$ and $p_1 - p_2 = 0.05$. Probability margins and entropy therefore help identify BLL–FSRQ boundary cases and other low-confidence candidates for follow-up prioritization.

Several limitations remain. The labeled sample is highly imbalanced, and some classes contain very few examples. Synthetic augmentation improves class balance but does not create new physical information. The feature set includes only catalog-level LAT observables, so multiwavelength data and contextual source information should further improve subclass separation, especially within the blazar population, as suggested by earlier AGN and blazar classification work that combines LAT information with broader source context (M. Ajello et al. 2020, 2022; H. Xiao et al. 2020; K. Cooper et al. 2023). The three spectral representations also remain phenomenological catalog fits rather than full physical source models. In addition, we do not test cross-catalog generalization or temporal robustness across different LAT catalog releases, so we leave the classifier’s stability under catalog evolution for future work. Our augmentation scheme is algorithmic rather than physically motivated, and its effect on rare-class boundaries deserves further study. Our decision to retain the singleton `gc` label improves catalog completeness at the prediction stage, but it does not yield a reliable class-specific performance estimate for that category.

These limitations suggest several directions for future work. Future work can combine the present spectral-type-aware framework with multiwavelength features, probabilistic calibration, or hierarchical class structures. It can also test whether agreement among PL, LP, and PLEC predictions provides a useful confidence criterion for follow-up prioritization. More broadly, our results show that spectral-type-aware multiclass learning offers a practical route toward refining the catalog-level subclass census of the unresolved *Fermi*-LAT sky.

5.2. Conclusions

We developed a fine-grained multiclass framework for classifying unknown 4FGL-DR4 γ -ray sources. We defined the unknown-source population as the union of blank-class sources, **unk** sources, and BCUs, and we treated the task as a 22-class supervised classification problem.

We represented each source in three parallel spectral forms, PL, LP, and PLEC, and we trained an independent classifier for each representation. This design allowed us to test how spectral parameterization affects classification performance and source-level assignments.

We compared TabNet, RF, and a stacked ensemble. The ensemble delivered the strongest and most stable performance across all three spectral representations. It reached accuracies of 0.9295, 0.9245, and 0.9413 and weighted F_1 scores of 0.9297, 0.9255, and 0.9416 for PL, LP, and PLEC, respectively. Among the three spectral representations, PLEC produced the best overall validation results, indicating that cutoff-sensitive spectral features provide the most informative representation for this multiclass problem.

The feature-set and uncertainty-ablation experiments identify the best final configuration as the uncertainty-aware PLEC model with $\lambda_{\max} = 0.2$, which reaches accuracy and weighted F_1 of 0.9614. The AllSpec uncertainty-aware configuration is also competitive, with accuracy 0.9581 and weighted $F_1 = 0.9582$, demonstrating that the simultaneous use of PL, LP, and PLEC parameters is a valid use of the full catalog spectral information. The photometry-plus-variability baseline remains strong, confirming that non-spectral temporal and broadband flux information is a major source of class discrimination.

We provide probabilistic classifications for all 4186 unresolved sources, including top-three class probabilities, full 22-class probability vectors, probability margins, and predictive entropy. These outputs are intended to support confidence-aware source selection, BCU subclassification, and follow-up prioritization.

The class-wise results showed that the classifier separated most source classes well. The dominant residual confusion occurred between BLL and FSRQ, which reflects the intrinsic overlap between the two main blazar classes in catalog-level LAT observables. In contrast, the model separated pulsar-related classes more cleanly, especially under the PLEC representation.

Our prediction catalogs provide catalog-level class assignments for the unresolved 4FGL-DR4 population and thereby offer physically motivated hypotheses for follow-up study. They can support population studies, counterpart searches, and prioritization of follow-up for unknown LAT sources. More generally, this work shows that spectral-type-aware machine learning provides a practical route toward improving fine-grained catalog-level classification of the unresolved γ -ray sky in a way that complements recent hierarchical, probabilistic, and direct multiclass approaches (A. Bhat & D. Malyshev 2022; S. Germani et al. 2021; D. Malyshev & A. Bhat 2023; D. V. Malyshev 2023; J. Coronado-Blázquez 2022).

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APPENDIX

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A. SUPPLEMENTARY FIGURES

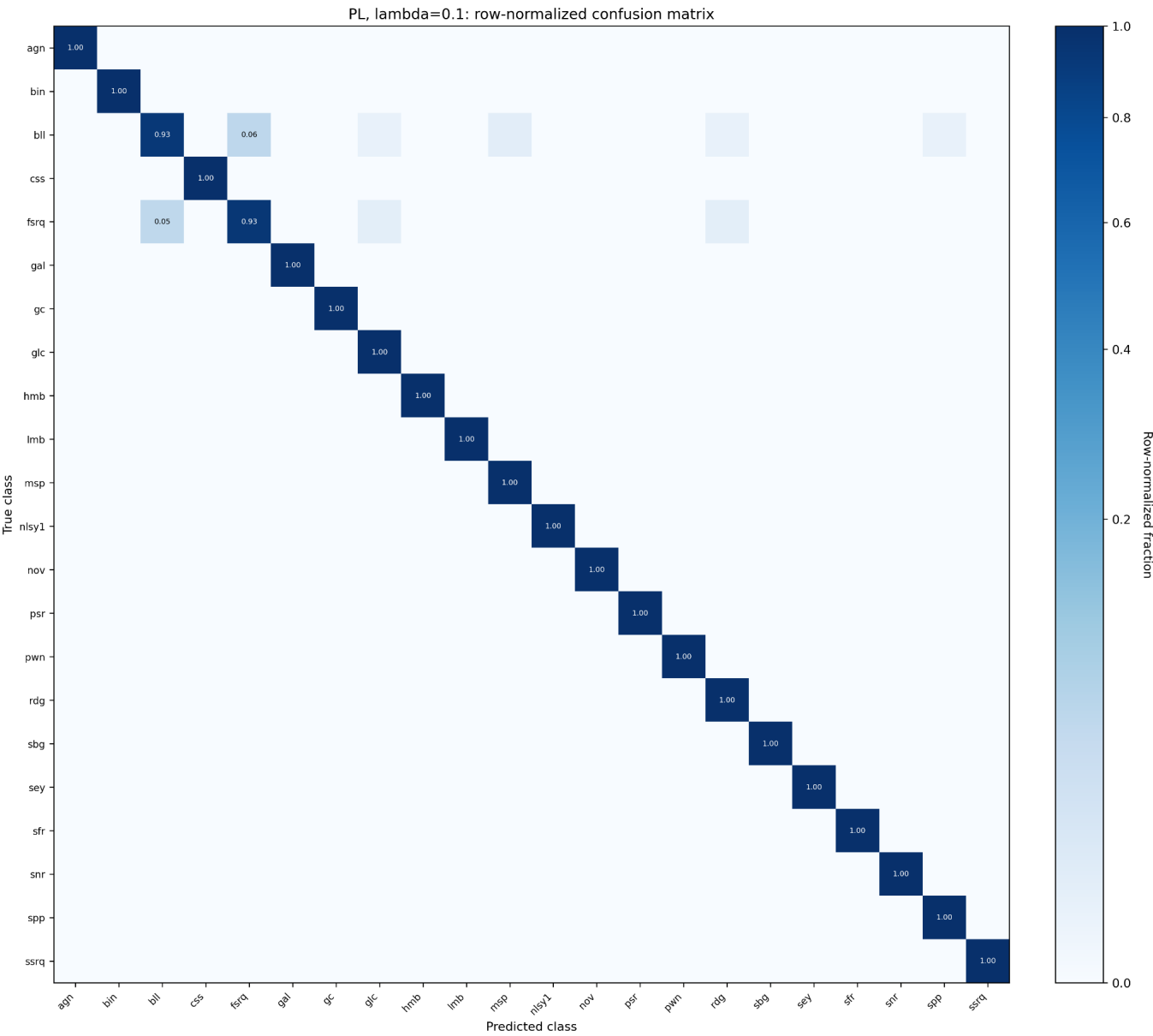


Figure 1. Row-normalized confusion matrix for the PL validation set.

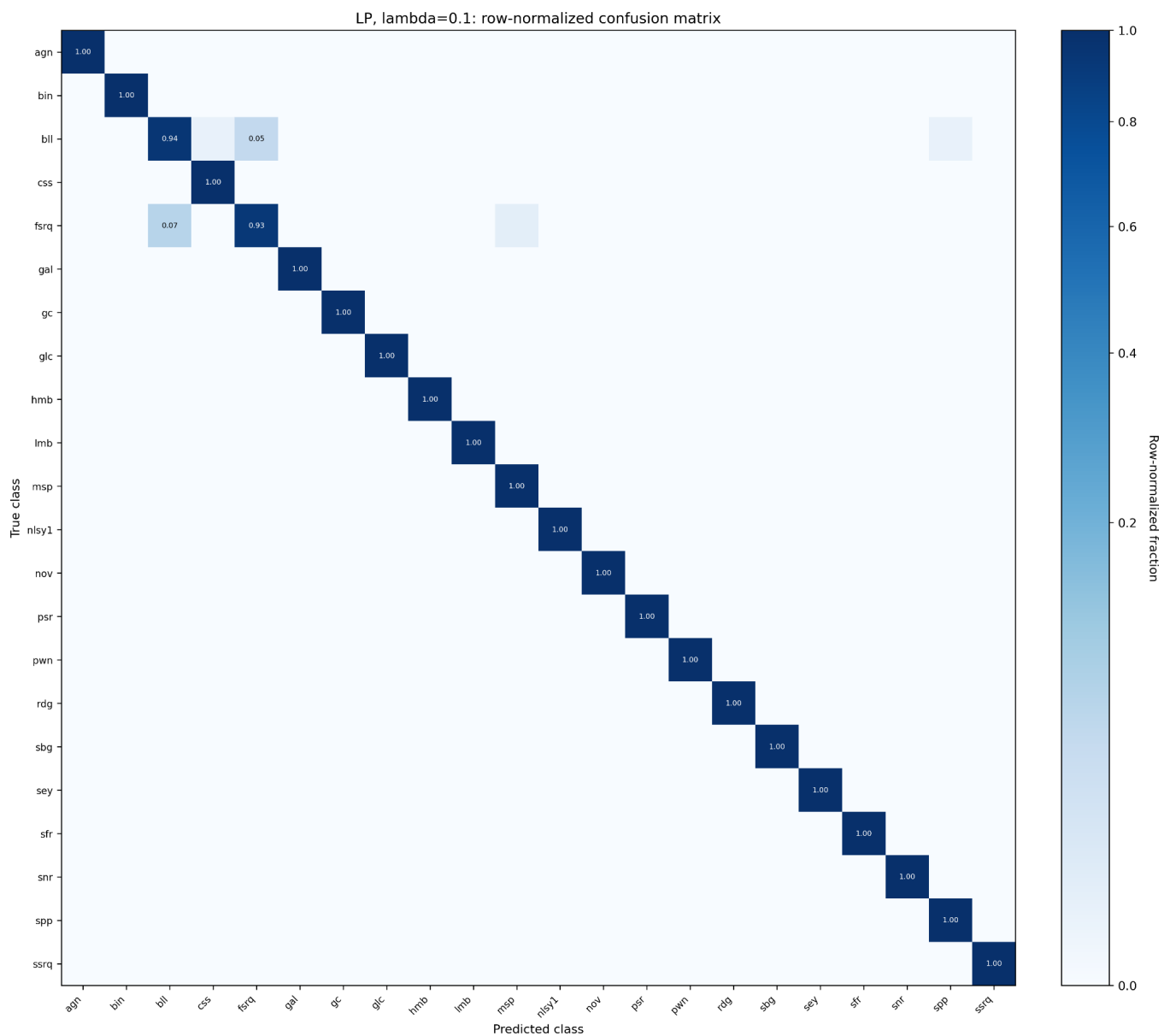


Figure 2. Row-normalized confusion matrix for the LP validation set.

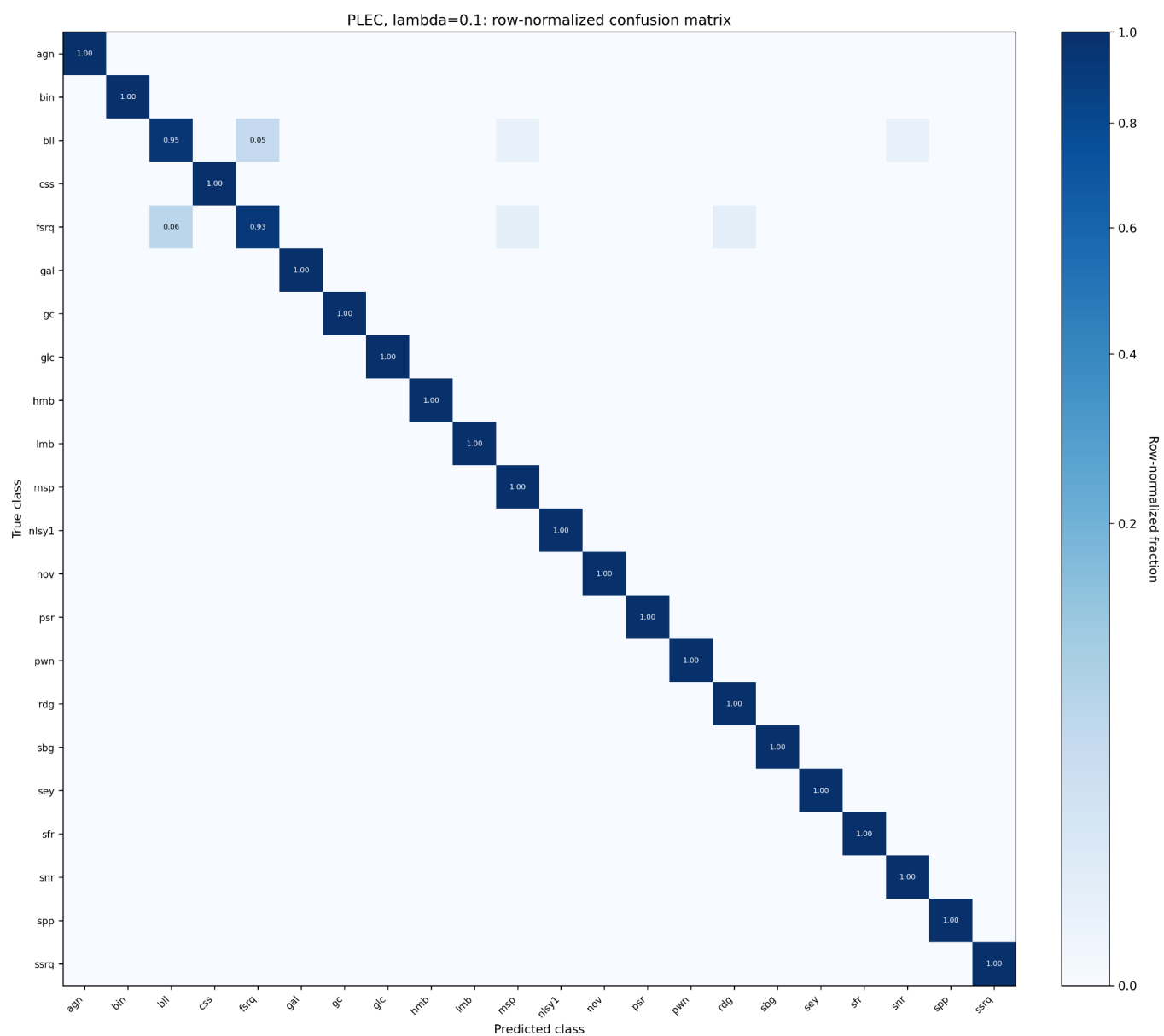


Figure 3. Row-normalized confusion matrix for the PLEC validation set.

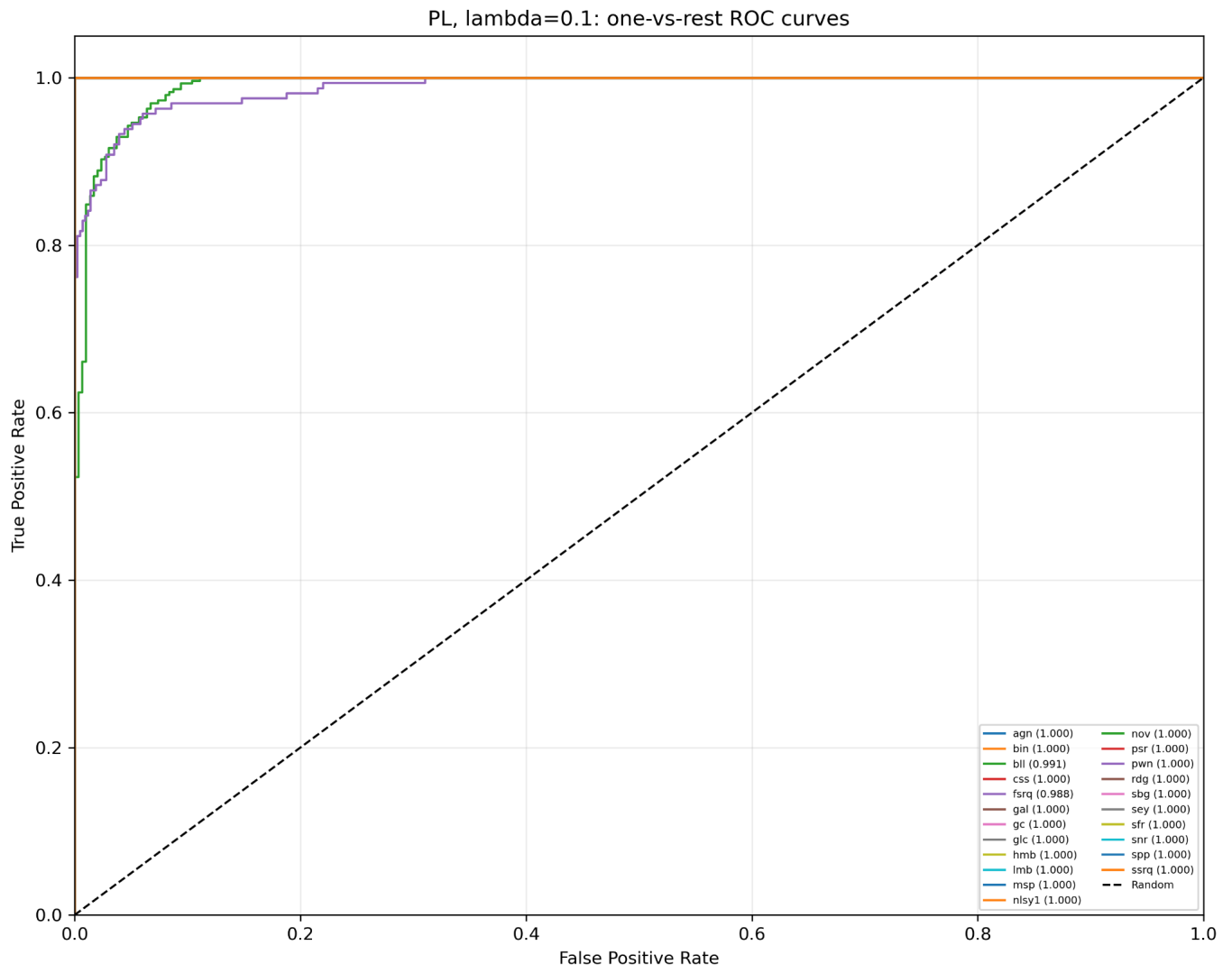


Figure 4. One-versus-rest ROC curves for the PL validation set.

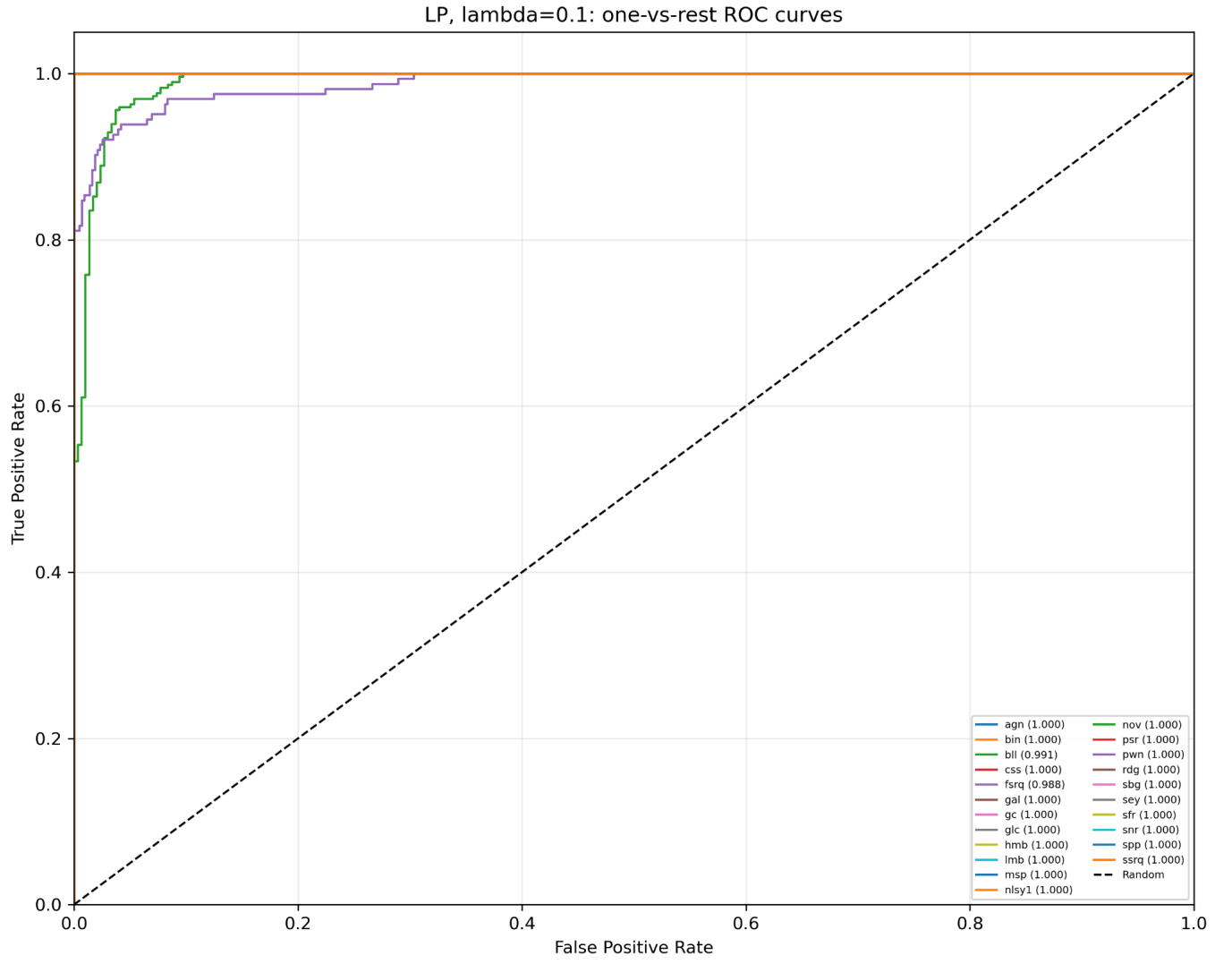


Figure 5. One-versus-rest ROC curves for the LP validation set.

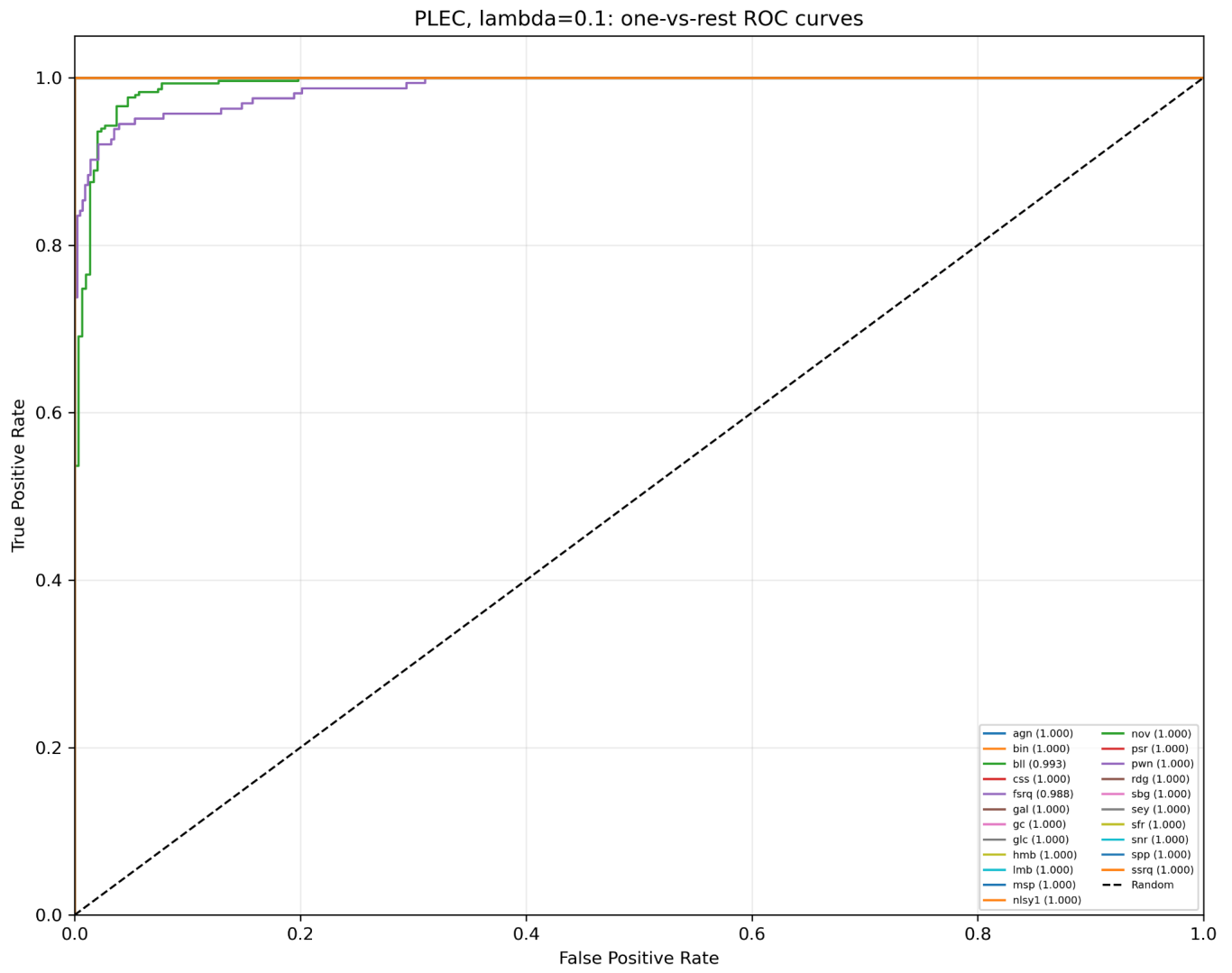


Figure 6. One-versus-rest ROC curves for the PLEC validation set.

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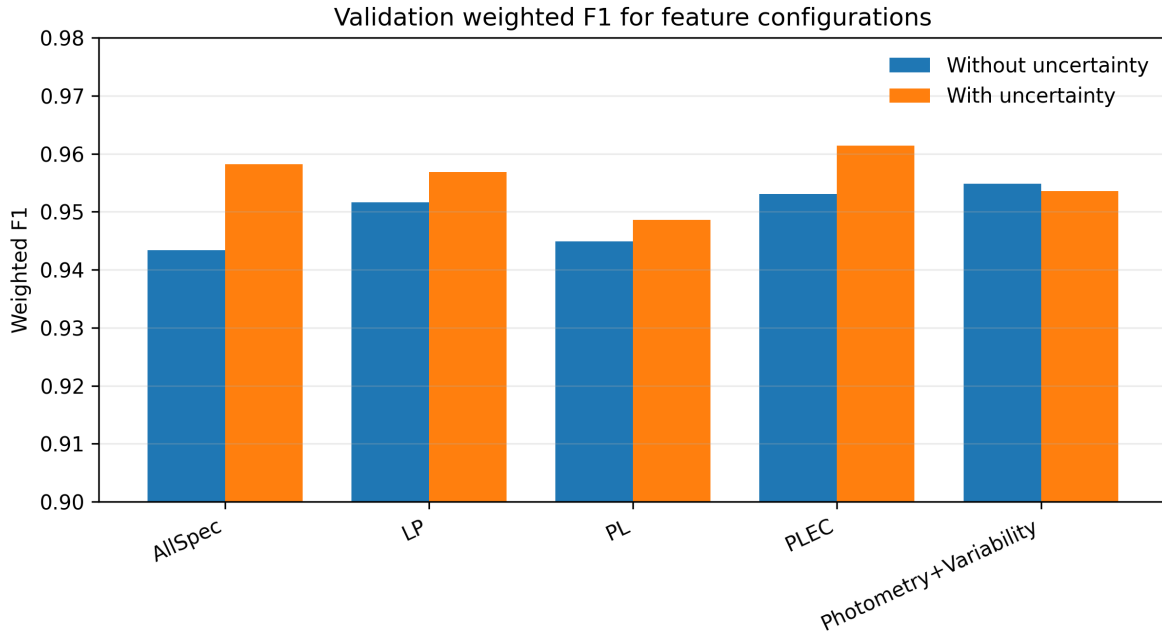


Figure 7. Validation weighted F_1 for the five feature configurations with and without uncertainty columns.

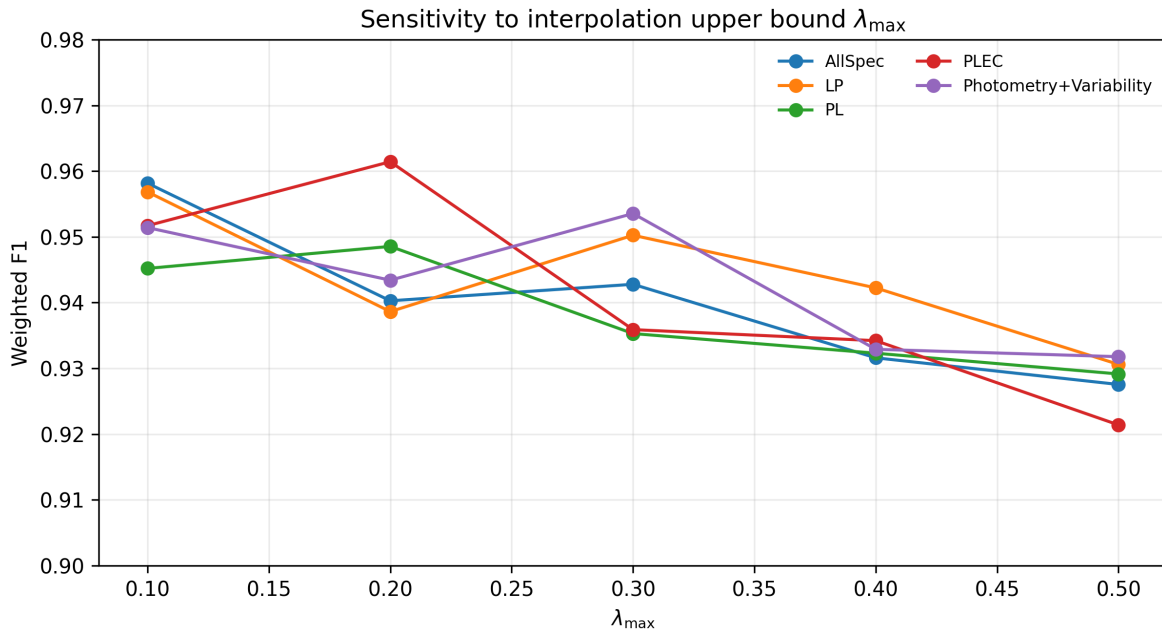


Figure 8. Weighted- F_1 sensitivity to the interpolation upper bound $\lambda_{\max}$ for the uncertainty-aware configurations.

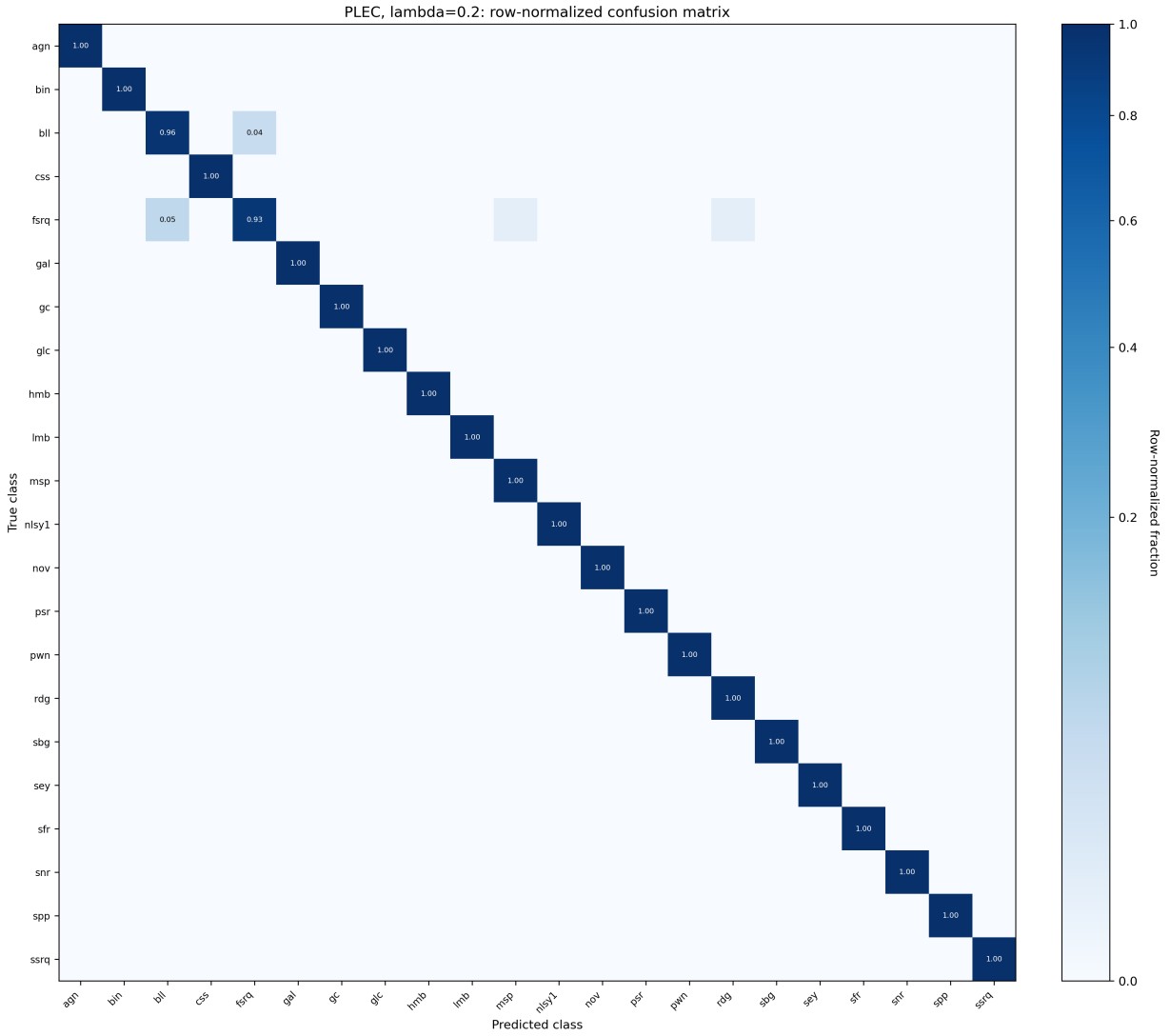


Figure 9. Row-normalized confusion matrix for the best PLEC uncertainty-aware model. A nonlinear color mapping is used to make small off-diagonal terms visible.

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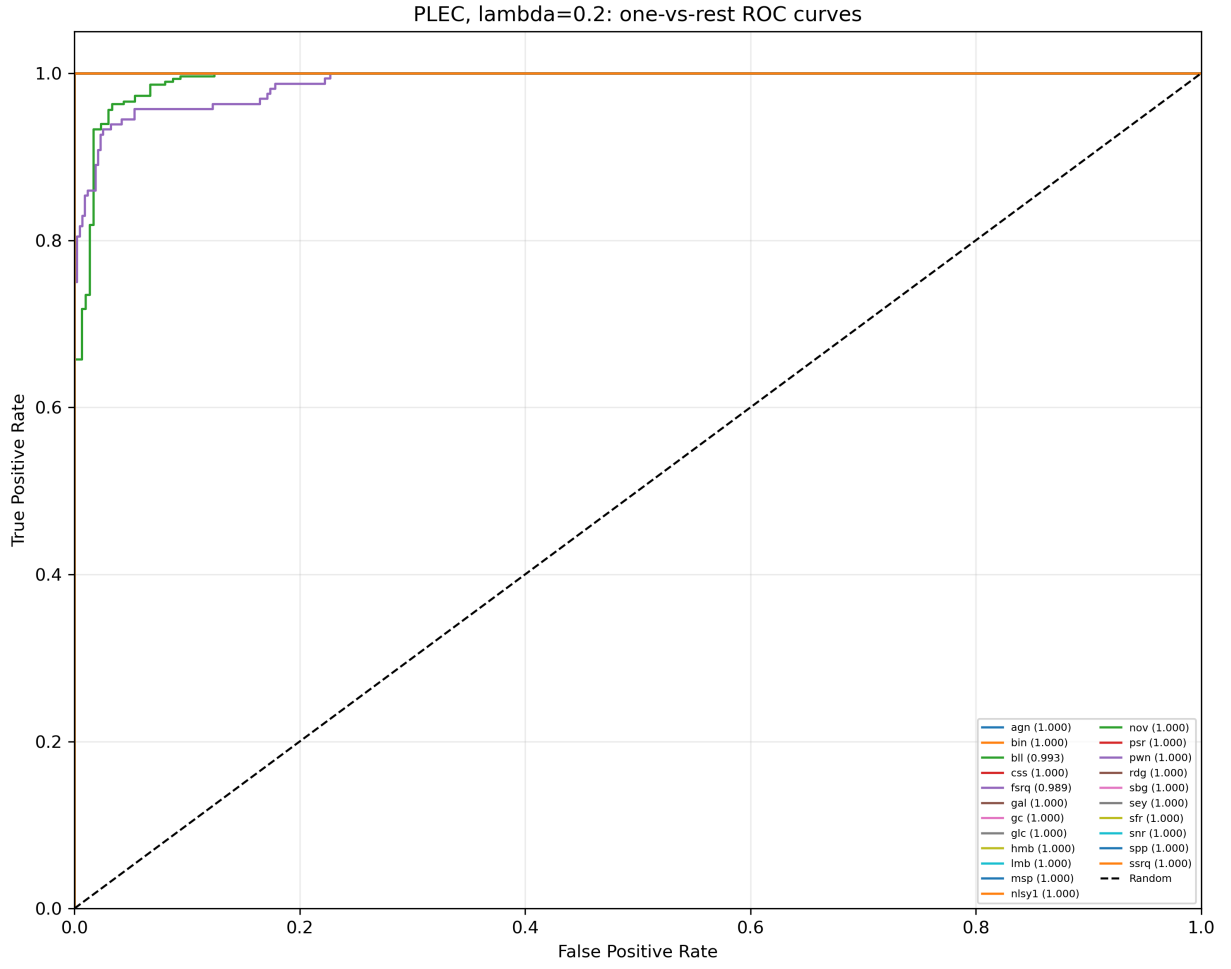


Figure 10. One-versus-rest ROC curves for the best PLEC uncertainty-aware model.

